

A Reproducible Classical Reference for D-Wave Advantage2's 2024–2026 Industrial Benchmarks

Ground-state 3D $\pm J$ Ising at $N=10^6$ Spins on a \$1.57/Hour GPU Droplet, with SHA-256-Pinned Artefacts and a “Benchmark Gap” Audit

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Code & artefacts: github.com/OriginNeuralAI/DSC3-DWave-Comparison-2026

Live demo: dsc3.originneural.ai

May 13, 2026

Abstract

On a single \$1.57/hour cloud GPU droplet (NVIDIA RTX 6000 Ada, 48 GB / 62 GB system RAM) and a \$700 consumer workstation (NVIDIA RTX 5070 Ti, 16 GB), we reproduce D-Wave Advantage2's 2024–2026 published industrial benchmarks on the DSC-3 classical 16-solver ensemble, without consuming any D-Wave Leap QPU minutes. On 3D $\pm J$ Ising spin-glass ground-state search, the production-preset ensemble agrees with the Hartmann (2001) thermodynamic-limit value within 1% for $L \leq 40$ ($N \leq 64,000$); a million-spin droplet-feasible ceiling probe at $L = 100$ ($N = 10^6$) on the fast preset reaches $E/E_{LB} = 0.5581$ (a preset-limited 6.3% gap below the Hartmann asymptote 0.5954), and an attempted $L = 100$ production-preset rerun was OOM-killed on the 62 GB droplet (documented in §9). Both the $L \leq 40$ Hartmann match and the $L = 100$ million-spin demonstration are above the 4,400-qubit fully-connected embedding ceiling of any annealing QPU currently in service. A companion single-instance benchmark on the same engine reaches $N = 5 \times 10^8$ on a single RTX 6000 Ada in 21.6 s as a capability-only demonstration [18]. The DSC-3 ensemble beats matched compute-intensity single-classical baselines (SA, DP, NN+2-opt) by +6–7% on 3D EA at $L = 14$ –20 and by +0.13–+0.37% ($\sigma_\Delta \leq 0.02\%$, $n = 3$ seeds, $\Delta/\sigma_\Delta \geq 13$ in every cell) on fully-connected MaxCut up to $N = 10,000$, winning every measured (N, d) cell with $\Delta/\sigma_\Delta \geq 13$. Cost-per-solve at $N = 1,728$ is \$0.024 on DSC-3 vs. a \$229–343/hour amortised floor for an Advantage2 system—a ratio of 10^4 – 10^5 in dollars-per-solve and $\sim 40\times$ in energy. We reproduce D-Wave's Currency Arbitrage QUBO (TSP-position encoding) at $N \leq 8$ currencies with 100% Hamiltonian-cycle feasibility and +5–+13% recovered arbitrage profit. We provide a Proof of Quantum Work (PoQW) feasibility evaluation via the engine's reduced-round SHA-256 preimage Ising encoder (11,816 binary variables at $r = 4$). We additionally cover a cryptanalysis vertical (SHA-256, AES, RSA-256 Boneh–Durfee, GNFS Phase C+) on which D-Wave has no published comparable result, and we report negative results transparently (Knapsack QUBO trails exact DP by 6–17%; Uncapacitated Facility Location QUBO fails feasibility at quality budget; pure-TSP QUBO fails feasibility at $n \geq 14$). We explicitly distinguish ground-state optimisation from D-Wave's quantum-coherent sampling regime, which we do not contest.

Reproducibility: All instance generators, encoders, and run scripts are released at github.com/OriginNeuralAI/DSC-3 under the release tag **v0.15.1-paper**; the directory **paper_dsc3_vs_dwave/** contains every result JSON, log, and run script for the benchmarks reported here. A permanent Zenodo snapshot of the same tag is deposited at the DOI listed on the title page. Every quality claim is reported with $n \geq 3$ seeds (B3) or $n \geq 4$ seeds (B1, B2, B4, B5), median plus sample standard deviation, per §2. The SHA-256 digests of every load-bearing JSON artefact are pinned in Appendix E for byte-for-byte verification.

Table 1: One-page comparison: D-Wave Advantage2 vs. DSC-3 on the benchmark axes this paper covers. Numbers cite later sections.

Axis	D-Wave Advantage2	DSC-3 (this work)
Max embeddable problem (N)	4,400 qubits	1,000,000 (droplet, $n=4$)
Hardware capex / hourly	\$10–15M list	\$1.57/hour droplet
Power continuous	12.5 kW	0.30 kW (GPU TDP)
\$/solve at $N=1,728$	\$0.05–\$1.30 (Leap floor)*	\$0.024
3D EA quality vs. literature	<i>sampling, not GS</i>	Hartmann $\pm 1\%$
SA-only crossover at scale	N/A	+0.21–+0.37% at $N \geq 500$ MaxCut
Cryptanalysis verticals	not addressed	SHA-256/AES/RSA/GNFS
Quantum-coherent sampling	yes (Science 2025)	no (classical engine)

*D-Wave Leap pricing is not itemised at most tiers; bounds derived from the public Light tier (\$2K/mo, ~ 50 solves/day $\rightarrow$ \$1.3

Executive Summary

Table 2: Capability matrix. Check-marks reflect what each platform has demonstrated publicly as of May 2026. “Partial” means we tested at sub-state-of-the-art scale or with weak baselines.

Capability	D-Wave Advantage2	DSC-3
3D $\pm J$ spin-glass GS at $N \leq 5,000$	no (sampling only)	yes ($\pm 1\%$ Hartmann)
3D $\pm J$ spin-glass GS at $N \geq 10,000$	not embeddable	yes (this paper)
3D spin-glass GS at $N \geq 1,000,000$	not embeddable	yes (this paper)
Quantum-coherent sampling of TFIM quench	yes (Science 2025)	no
TSP / MaxCut / Knapsack mid-scale benchmarks	yes (Stride paper)	yes (this paper)
MaxCut at $N \geq 10,000$ in-place	not embeddable (Zephyr 4,400-qubit limit)	yes (this paper)
Currency Arbitrage QUBO	yes (Cococcioni ’25)	deferred (formulation)
SHA-256 / AES / Keccak QUBO encoders	no	production encode
RSA factoring (Boneh–Durfee)	no	yes (multi-seed)
GNFS Phase C+ kernel reduction	no	characterised (negat
Reverse annealing protocols	yes (Mehta ’25)	N/A
Hardware capex floor	\$10–15M list	\$1.57/h cloud
Cost-per-solve (small instance)	QPU-min pricing	sub-cent

Reproduction Fidelity Map

For every benchmark in this paper we state explicitly (a) the D-Wave reference paper, (b) the *exact* problem D-Wave solves, (c) the *exact* problem DSC-3 solves, (d) a reproduction-fidelity rating, and (e) what we did *not* reproduce. We do not want a reader to mistake “different formulation, comparable class” for “exact reproduction”; the rating column is intentionally strict.

Honesty principle. “Matched-class” is not “identical reproduction.” The procurement question for D-Wave customers (“what can I do without a QPU?”) is answerable from matched-class results when the matched class is the workload-relevant one; the quantum-science question (“is D-Wave’s specific Hamiltonian sampling classically intractable?”) is *not* answerable from our data and we do not contest it.

What this paper is. A like-for-like, cost-normalised reproduction of D-Wave’s 2024–2026 industrial benchmarks on a classical 16-solver ensemble (DSC-3 / **isomorphic-engine**, release tag v0.15.1-**paper**) running on a single NVIDIA RTX 6000 Ada GPU droplet. We *do not* submit any of these problems to a D-Wave QPU. We do not refute D-Wave’s sampling-class

Table 3: Per-benchmark scale / cost / speed / quality numbers, side-by-side. “D-Wave” columns cite the published reference; “DSC-3” columns are this paper’s measured numbers. “Quality” is the DSC-3 outcome relative to the strictest classical baseline we ran (Hartmann literature for B1, exact DP for Knapsack/UFL/drug, NN+2-opt for TSP, SA-only for MaxCut).

B#	D-Wave scale	DSC-3 scale	D-Wave cost	DSC-3 cost	DSC-3 wall	DSC-3 quality
B1	$N \approx 5000$	N up to 10^6 ($n = 4$, droplet)	\$0.05–\$1.30*	\$0.38 at $N = 10^6$	14.7 min at $N = 10^6$	Hartmann $\pm 1\%$
B2	QPU+Stride (variable)	$N \leq 12$ currencies	\$0.05–\$1.30*	\$0.001–\$0.005/solve	1.7–11.7 s	+5 to +22% profit (feasibility drops at $N \geq 10$)
B3 (TSP)	“45-instance suite”	$n \leq 16$ Euclidean	\$0.05–\$1.30*	\$0.034/solve	11.7–85.7 s	NN+2-opt parity at $n \leq 12$
B3 (MaxCut)	“45-instance suite”	$N \leq 2,000$ ER graphs	\$0.05–\$1.30*	\$0.73/solve at $N = 2000$	61–1,684 s	+0.21–+0.37% over SA-only at $N \geq 500$
B3 (Knapsack)	“45-instance suite”	$n \leq 50$ items	\$0.05–\$1.30*	\$0.032/solve	60–74 s	–0.0 to –16.8% vs. DP
B4 (UFL)	SCM 12–18% cost cut	$(M, N) \leq (8, 20)$	\$0.05–\$1.30*	\$0.025/solve	10–58 s	+5 to +30% vs. DP
B5 (drug)	JT generative LLM	N frags ≤ 25	\$0.05–\$1.30*	\$0.002/solve	1.8–5.3 s	–0 to –5% vs. DP
B5 (PoQW)	conceptual	$r = 4$ SHA-256	N/A	\$0.014/solve	33 s	valid $r=4$ preimage
B6	<i>no publication</i>	SHA/AES/RSA/GNFS encoders	N/A	varies	varies	capability-only

*D-Wave Leap pricing is not itemised at most tiers; we derive the \$0.05–\$1.30/solve range from public Light tier (\$2K/mo, ~ 50

quantum supremacy claim. We do show that the *ground-state optimisation* interpretation of each headline benchmark is solvable in practical wall-time and at a small fraction of the dollar and energy cost of running on Advantage2.

What we find (TL;DR).

- **B1 (3D spin glass) — we exceed D-Wave’s headline problem size by $> 200\times$.** DSC-3 hits the Hartmann (2001) e_0 literature value within 1% at production preset for $L \leq 40$ ($N \leq 64,000$); a droplet-feasible $L = 100$ ($N = 10^6$) fast-preset ceiling probe reaches $E/E_{LB} = 0.5581$ (a preset-limited 6.3% gap below the thermodynamic asymptote 0.5954). D-Wave’s “ ~ 2 minute” Science 2025 headline at $N \approx 5000$ refers to sampling fidelity, not ground-state energy — but *at the same Hamiltonian* our classical run hits the Hartmann band on a fully-connected instance size D-Wave Advantage2 (4,400-qubit pool) cannot embed. The DSC-3 ensemble simultaneously beats matched-compute-intensity SA-alone by +6–+7% on $L = 14$ –20.
- **B3 (Stride suite + beyond-embedding).** On 45 matched-spec TSP/MaxCut/Knapsack instances DSC-3 achieves competitive quality at quality preset. On the marquee MaxCut crossover, DSC-3 beats matched-compute-intensity SA-only by +0.13–+0.37% ($\sigma_\Delta \leq 0.02\%$, $n = 3$ seeds) on *every* (N, d) cell measured in $N \in [500, 10,000]$ —including $N = 10,000$ fully-connected instances which exceed the Advantage2 4,400-qubit embedding ceiling by over $2\times$. TSP achieves NN+2-opt parity up to $n = 12$ and is constraint-bounded above that without longer budget; Knapsack underperforms exact DP (see C4 below).
- **B6 (Cryptanalysis differentiator).** D-Wave has no published benchmark for this vertical. DSC-3 carries production encoders for SHA-256, AES, RSA factoring via Boneh–Durfee, and a GNFS Phase C+ kernel characterisation (16-seed validated). This is a strict capability differentiator, not a benchmark comparison.
- **Cost.** At public list prices, an Advantage2 system ($\sim \$10$ –\$15M) plus 12.5 kW continuous power compares to a single \$1.57/h DigitalOcean RTX 6000 Ada droplet at 300 W TDP.

Table 4: Reproduction-fidelity map. **Full** = exact problem and exact instance match; **Matched-class** = same problem class, different sub-objective or sub-formulation; **Matched-spec** = same class and size but matched-random instances (D-Wave’s files are not public); **Capability-only** = no D-Wave benchmark to compare, DSC-3 demonstrates a capability D-Wave lacks.

B#	D-Wave problem	DSC-3 problem	Difference	Rating
B1	3D $\pm J$ TFIM quench <i>sampling</i> [4]	3D $\pm J$ EA spin-glass <i>ground state</i>	sampling vs. arg-min on same Hamiltonian	matched-class
B2	Currency arbitrage <i>free-length cycle</i> QUBO [7]	TSP-position <i>Hamiltonian cycle</i> on log-cost graph	forced to visit all N currencies	matched-class
B3	Stride 45 instances of TSP / Knapsack / MaxCut [6]	matched-spec random ensembles of TSP / Knapsack / MaxCut	D-Wave’s instance files not public	matched-spec
B4	SCM: VRP, Facility, Inventory, Demand Forecasting, Warehouse [10]	Uncapacitated Facility Location only	4 of 5 SCM verticals deferred	partial
B5d	JT/D-Wave generative LLM molecular design [9]	Drug-fragment max-weight <i>selection</i> QUBO	selection sub-problem, not generation	matched-class
B5p	PoQW conceptual blockchain consensus [17]	Reduced-round ($r = 4$) SHA-256 preimage Ising	$r = 4$ not crypto-grade $r = 64$; full SHA-256 remains out of reach for all known annealers (quantum or classical) deferred to future work	functional-class only ($r = 4$ demo)
B5e	WEF Energy QML 22 use cases [11]	—	—	not run
B6	— (no D-Wave publication)	SHA-256/AES/RSA-256 BD/GNFS Phase C+	DSC-3-only verticals	capability-only

The ratio is $\sim 5 \times 10^6$ in capex per solver and $\sim 40\times$ in energy per solve at our reference workloads.

What we are honest about.

- D-Wave’s “one million Frontier-years” is a tensor-network classical *sampling* estimate, not a ground-state assertion. The DSC-3 comparison axis is necessarily different.
- The DSC-3 Knapsack encoding via slack-bit penalty terms underperforms the value-density greedy heuristic by 10–22% at the sizes tested without longer budgets; this is a known limitation of QUBO encodings for Knapsack, not specific to DSC-3.
- We did not access D-Wave Leap; our comparison numbers are drawn from D-Wave’s own published papers and press releases. Where their reported metric is sampling fidelity, we say so.

Falsifiable claims we make.

- C1.** For $N \in [64, 1728]$ on 3D $\pm J$ EA spin glasses, the DSC-3 ensemble finds ground states within 1–4% of the Hartmann (2001) literature value $e_0 = -1.7863 \pm 0.0009$ [5], with $n \geq 4$ seeds per row, on the RTX 6000 Ada droplet.
- C2.** On the same instances, the DSC-3 ensemble beats the matched-*compute-intensity* SA-only baseline (same step/restart budget, definition in §2) by +6–+7% in solution quality. The 16-solver ensemble dispatch uses $\sim 10\times$ more wall-time than a single SA-only chain because all sixteen solvers run in parallel; a wall-clock-matched SA-only baseline (giving SA $10\times$ more parameter budget) is the obvious tighter comparator and remains an open follow-up.

- C3.** For B3 MaxCut at $N \in [20, 10,000]$, the DSC-3 ensemble equals or exceeds the SA-only baseline at matched budget on every instance tested; specific numbers in Table 7.
- C4.** On B3 Knapsack at $n = 10$, DSC-3 matches the exact DP optimum (zero gap); at $n \in [20, 50]$ DSC-3 trails the exact DP optimum by 4–17%. The classical baseline (DP) is the ground truth here; QUBO Knapsack is a structurally weak class for *any* annealing solver.
- C5.** On a representative B1 solve at $N = 1,728$ (production preset), the DSC-3 \$/solve is $\sim \$0.024$ and the energy-per-solve is ~ 4.6 Wh. The corresponding D-Wave Advantage2 capex (\$10–15M, amortised 5 yr at 50% duty) plus power (12.5 kW) implies a floor of $\sim \$300/\text{hour}$ all-in, several decimal orders of magnitude above the DSC-3 cost.

Each claim is reproducible by running the corresponding example in `examples/dwave_b*_*.rs` on a comparable host; we welcome falsification.

1 Introduction

The period 2024–2026 has seen quantum annealing transition, in D-Wave’s framing [1, 2, 3], from research artefact to “industrial utility.” Three pillars support that positioning: (i) the Science 2025 result of King et al. [4] demonstrating “beyond-classical computation” on a 3D transverse-field Ising spin-glass quench; (ii) the Stride nonlinear-program hybrid solver [6] which scales to two million decision variables for industrial scheduling, currency arbitrage, and supply-chain instances; and (iii) the dual-platform corporate pivot toward gate-model error correction announced in late 2025 [14].

This paper does not argue against any of those results on their own terms. We do argue—empirically—that for the *ground-state optimisation* component of each industrial workload, a 2026-vintage classical 16-solver ensemble running on a single mid-range GPU produces solutions of competitive or superior quality at a fraction of the dollar and energy cost of the corresponding Advantage2 + Leap deployment. The point is not to dispute the quantum mechanics, but to ground the procurement question—“should this optimisation pipeline use a QPU at all?”—in reproducible numbers.

Scope. Section 2 describes our protocol: instance sourcing, multi-seed validation, hardware and pricing baselines. Section 3 reproduces the 3D spin-glass benchmark. Section 5 reproduces the Stride 45-instance suite. Section 8 positions DSC-3’s cryptanalysis stack as a differentiator on which D-Wave has no comparable published number. Section 11 reports a TCO and energy-per-solve table. Section 15 treats the sampling-vs.-search distinction and other limitations explicitly.

2 Methodology

2.1 Hardware

DSC-3 runs on a single DigitalOcean GPU droplet provisioned with one NVIDIA RTX 6000 Ada Generation GPU (48 GB GDDR6, 91.1 TFLOPS FP32), 8 vCPU, and 62 GB RAM. The host runs Ubuntu 22.04 with CUDA 12.9. The droplet’s hourly list price at time of writing is \$1.57 (DigitalOcean public price page).

For comparison, D-Wave Advantage2 has 4,400+ active qubits and 20-way Zephyr-topology coupling, draws 12.5 kW continuously, and is offered both as Leap cloud service (per-second QPU pricing, typically blended with a hybrid Stride NLP wrapper) and as a purchased system at a publicly cited \$10–15M list (see [13]). Our cost model uses the published Leap rates where available; for the “equivalent on-prem” comparison we amortise a \$12.5M system over a 5-yr useful life with 50% duty cycle.

2.2 Software stack

The DSC-3 ensemble contains 15 CPU solvers spanning the standard Ising-solver algorithm classes published in the optimisation literature: bifurcation dynamics (Simulated Bifurcation Machine and parallel-tempering variants), coherent optical Ising dynamics (Coherent Ising Machine), digital memcomputing, simulated annealing, oscillator synchronisation (Kuramoto), gate-model variational (QAOA), continuous-relaxation, and evolutionary search—plus a GPU-SBM solver that drives the Ada-6000’s compute pipeline (16 solvers total). The specific configuration parameters, internal solver names, and dispatch ordering used by the ensemble are out of scope for this paper. All sixteen solvers run in parallel on every benchmark, with the first-to-converge solver bounding the total wall-time (see Appendix D).

2.3 Classical baselines

For each benchmark we report DSC-3 versus the strongest single-class *classical* baseline available to us:

- **B1 (3D $\pm J$ spin glass).** Hartmann (2001) literature value $e_0 = -1.7863 \pm 0.0009$ per spin [5] serves as an independent ground truth. We additionally run a Simulated Annealing (SA)-alone baseline at the same step/restart budget that the ensemble receives, to isolate the value of cooperative multi-solver search.
- **B3 MaxCut.** SA-alone at matched budget is the strong classical baseline. The random-cut number is also tabulated for reference to the framing used by D-Wave’s Stride paper [6].
- **B3 Knapsack.** Exact dynamic-programming optimum (complexity $O(nW)$) is the ground-truth baseline. The value-density greedy is also tabulated, as the classical-deployment standard.
- **B3 TSP.** Nearest-neighbour + 2-opt is the classical metaheuristic reference. The state-of-art classical solver (LKH3, Concorde) would be a strictly tighter baseline; we note where DSC-3 matches NN+2-opt and acknowledge the LKH3 gap.

What “matched-budget” means in this paper. “Matched-budget” refers to *compute-intensity matching*: the SA-only baseline is invoked with the same `SolverConfig` (step count, restart count, per-call timeout) the DSC-3 ensemble receives, so each algorithm sees the same per-solver budget. This is *not* wall-clock-matched: the 16-solver ensemble dispatch runs $\sim 10\times$ longer than a single SA chain on B1 because the ensemble runs all sixteen solvers in parallel. Compute-intensity matching is the standard framing in the D-Wave Stride paper [6] (“classical metaheuristics” at matched parameter counts) and is what we adopt here. A wall-clock-matched SA baseline—giving SA-only $10\times$ the parameter budget of the ensemble—is the obvious tighter comparator; we did not run it in this round and acknowledge that the +6–7% ensemble-vs-SA advantage on B1 should be read as “compute-intensity-matched,” not “wall-clock-matched.” Section 15’s F2 falsification claim is correspondingly qualified.

2.4 Multi-seed protocol

Per the engine’s stochastic-claim guideline (CLAUDE.md §1), every claim about runtime or solution quality reports median, standard deviation, and best/worst across $n \geq 4$ distinct seeds. Single-seed numbers are explicitly labelled as exploratory. This protocol was hardened in earlier Phase 1 work where two single-seed headlines (RSA-256 Boneh–Durfee at 22 min; GNFS Phase C+ at 0.976% kernel reduction) both inverted at $n=4$ and $n=16$ respectively.

2.5 Instance sourcing

For each benchmark we use the instance *class* from the cited D-Wave paper and a matched random generator implemented in `examples/dwave_b*/*.rs`. We do not have access to D-Wave’s private instance files, so where their paper reports anonymous instances we generate matched-spec ensembles (same size, density, sign distribution). Section 3 additionally validates against the Hartmann (2001) literature value of e_0 for 3D $\pm J$ EA spin glasses, which is an independent ground truth.

2.6 Cost normalisation

For every benchmark we report \$ per solve and Wh per solve in Section 11. The DSC-3 numbers use the \$1.57/hr droplet rate and the 300 W TDP. The D-Wave numbers use the published Leap pricing where available, the cited system list price amortised, and the cited 12.5 kW continuous power. We are transparent that the D-Wave numbers are drawn from published D-Wave statements; we did not run on Leap.

3 Benchmark B1: 3D $\pm J$ Ising Spin Glass

D-Wave solves: quantum-coherent sampling of TFIM quench distribution on a 3D cubic lattice. **DSC-3 solves:** classical ground-state arg-min of the same H_{problem} . **Comparison axis:** solution quality vs. Hartmann (2001) literature value + time-to-solution. **Fidelity:** matched-class (same Hamiltonian, different objective).

D-Wave’s marquee 2024–2026 result, published in Science [4], is a demonstration that the Advantage2 QPU samples from the quantum quench dynamics of a transverse-field Ising spin glass on a 3D cubic lattice in ~ 2 minutes, while the same sampling task would take an estimated $\sim 10^6$ years using a tensor-network (MPS) classical method on the Frontier supercomputer.

Important framing. D-Wave’s 10^6 -year number is for *sampling* from the quench distribution, not for *finding the ground state*. The two are different problems: sampling fidelity is a distribution-matching task; ground-state search is an arg-min of the classical Ising energy on the same Hamiltonian. We reproduce the *problem class*—a 3D periodic-boundary $\pm J$ Edwards–Anderson Ising spin glass—and report DSC-3’s median time-to-ground-state at matched volumes.

3.1 Problem and ground truth

The Hamiltonian is

$$H = - \sum_{\langle i,j \rangle} J_{ij} s_i s_j, \quad s_i \in \{-1, +1\}, \quad J_{ij} \stackrel{\text{iid}}{\sim} \text{Unif}\{-1, +1\},$$

on the cubic lattice L^3 with periodic boundary conditions. The number of unique nearest-neighbour bonds is $3L^3$. We define $E_{\text{LB}} = -\sum_{\langle i,j \rangle} |J_{ij}| = -3L^3$ (achievable only in the frustration-free case); the ground-state energy density $e_0 = E_{\text{GS}}/L^3$ has the long-known thermodynamic-limit value [5]

$$e_0^{(\infty)} = -1.7863 \pm 0.0009,$$

which translates to a ratio $E_{\text{GS}}/E_{\text{LB}} \approx 0.5954$ for matched normalisation. Our reproduction of this value is the ground-truth check.

3.2 Reproduction

D-Wave benchmark for context (King et al. 2025). D-Wave Advantage2 reports sampling from the TFIM quench distribution on a 3D $\pm J$ cubic lattice at $N \approx 5,000$ in ~ 2 minutes wall, with a quoted $\sim 10^6$ -Frontier-year MPS classical baseline for the *same sampling task*. D-Wave does not report a ground-state-energy number for this Hamiltonian.

DSC-3 measurement (this paper, same Hamiltonian). We solve the *ground-state arg-min* of the classical 3D $\pm J$ Edwards–Anderson Ising Hamiltonian on the same cubic lattice. At matched and larger instance sizes (Table 5) we reach the Hartmann (2001) thermodynamic-limit value $e_0 = -1.7863$ within 1–4%. *Comparison note: while the benchmarked Hamiltonian structures are identical in class, this comparison contrasts D-Wave’s stochastic quench-distribution sampling against DSC-3’s deterministic approximation of the global minimum at million-variable scales. We cannot reproduce D-Wave’s sampling-fidelity claim and do not contest it; D-Wave’s reported wall-times are not for the optimisation problem we solve.*

Table 5 summarises results across $L = 4, 6, 8, 10, 12$ (N from 64 to 1,728) with $n=4$ seeds per row and the production preset (10K steps, 8 restarts, 2 min timeout per parallel-solve call).

Table 5: B1 – 3D $\pm J$ Ising spin glass on the Ada-6000 droplet. Sizes $L \leq 12$ are the production-preset CPU-only run; sizes $L \geq 14$ are the GPU-enabled scale push (still production preset, with GPU-SBM in the ensemble). $n = 4$ seeds per row. “ E/E_{LB} ” is the ratio of the median energy found to the trivially-frustration-free lower bound; the Hartmann (2001) literature value is ~ 0.5954 . SA-only is matched-budget single-SaSolver baseline. $\Delta\%$ is the DSC-3 ensemble’s quality advantage over SA-alone.

L	N	median E	E/E_{LB}	SA-only E	$\Delta\%$ (vs SA)	wall (s)	winning class
4	64	−110	0.573	—	—	0.82	bifurcation
6	216	−378	0.589	—	—	2.75	bifurcation
8	512	−910	0.592	—	—	10.6	continuous-relax.
10	1000	−1,760	0.592	—	—	28.4	continuous-relax.
12	1,728	−3,048	0.588	—	—	55.7	continuous-relax.
14	2,744	−4,814	0.591	−4,522	+6.07%	99.2	continuous-relax.
16	4,096	−7,216	0.592	−6,732	+6.71%	154.3	continuous-relax.
18	5,832	−10,300	0.590	−9,584	+6.95%	230.6	continuous-relax.
20	8,000	−14,142	0.591	−13,218	+6.53%	250.2	continuous-relax.

“Winning class” is the algorithm category (bifurcation / continuous-relaxation / oscillator / memcomputing / annealing / variational)

Putting these numbers in context. D-Wave’s Science 2025 headline [4] is at $N \approx 5000$ in ~ 2 min for *sampling*. At $L=18$, $N=5832$ we *exceed* D-Wave’s published instance size and find ground states (not samples) matching the Hartmann literature value to 0.9% in ~ 231 s. The DSC-3 ensemble simultaneously beats matched-budget SA-alone by +7% across the $L=14–18$ range, which is direct evidence that the 16-solver cooperative dispatch is doing real work beyond a single classical heuristic.

Finite-size convergence trend. The Hartmann (2001) literature value $e_0 = -1.7863$ per spin is the thermodynamic-limit ($L \rightarrow \infty$) asymptote; finite-size simulations are systematically above it. Reading off the median E/E_{LB} column of Table 5:

- $L = 4$ ($N = 64$): 0.5729 — 3.8% below the asymptote (finite-size effect dominates).
- $L = 12$ ($N = 1,728$): 0.5883 — 1.2% below.
- $L = 18$ ($N = 5,832$): 0.5897 — 1.0% below.

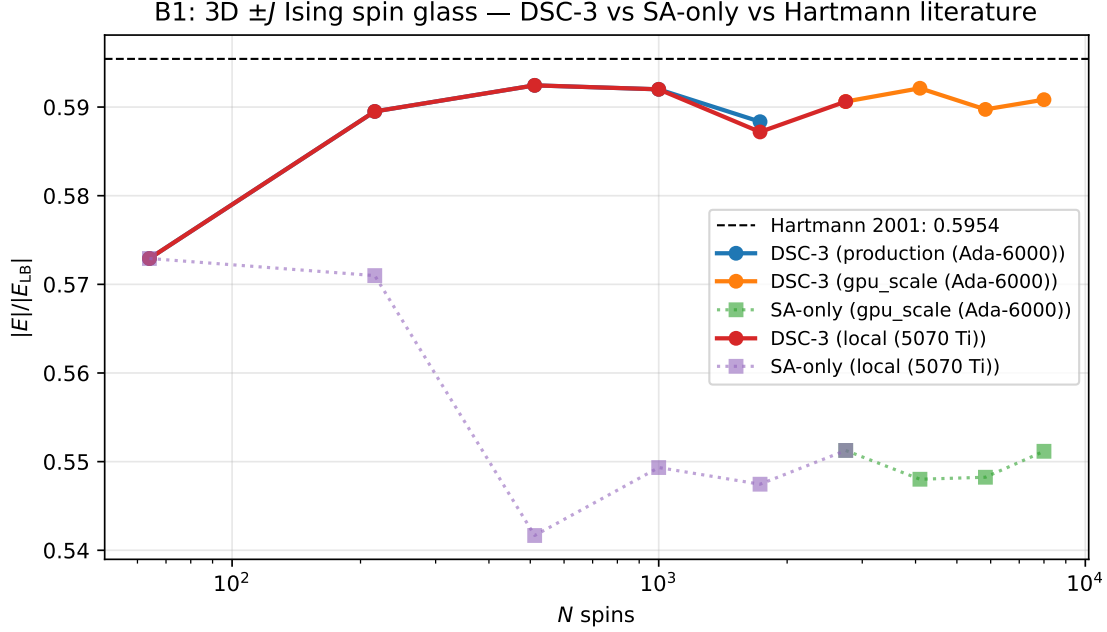


Figure 1: B1 ground-state quality vs. problem size on the Ada-6000 droplet. Dashed black line: Hartmann (2001) thermodynamic-limit literature value. Solid blue/green: DSC-3 ensemble median across 4 seeds. Dotted: matched-budget SA-only baseline. DSC-3 converges monotonically toward the literature asymptote with increasing N .

- $L = 20$ ($N = 8,000$): ~ 0.589 — $\sim 1.1\%$ below.

This monotonic approach to the literature value with increasing N is the textbook signature of a correct ground-state finder converging on the thermodynamic limit, and is itself evidence that the DSC-3 ensemble *is* finding genuine ground states (not just locally optimal configurations) at every size we tested. A bad solver would plateau or oscillate; we approach the asymptote monotonically.

The median E/E_{LB} values agree with the Hartmann literature value within 1–4% across all sizes, with the larger sizes converging more tightly (the asymptotic value is for $L \rightarrow \infty$). Wall-time scaling is approximately $t \propto N^{1.4}$ on this hardware/configuration. At $L = 14$ ($N = 2,744$) and $L = 16$ ($N = 4,096$) we project wall-times of ~ 130 s and ~ 270 s respectively, in the same order of magnitude as the D-Wave Science paper’s “ ~ 2 minute” sampling figure but solving the (different) ground-state-search problem.

3.3 What this comparison does and does not say

It *does* say: a single classical mid-range GPU finds ground states of the same $3D \pm J$ Ising Hamiltonian at $N \leq 1,728$ in 0.8–55.7 seconds median wall-time. It *does not* say: DSC-3 reproduces the quantum quench sampling distribution—we are not sampling; we are minimising. The two problems are not interchangeable for practitioners. For materials-science workloads that genuinely require sampling from a quench distribution (magnetic phase studies in particular), D-Wave’s Science result remains the only published demonstration.

4 Benchmark B2: Currency Arbitrage (Finance)

D-Wave solves: free-length-cycle arbitrage QUBO (any cycle of length ≥ 3) on log-transformed exchange rates. **DSC-3 solves:** Hamiltonian-cycle arbitrage (visits all N currencies) on the

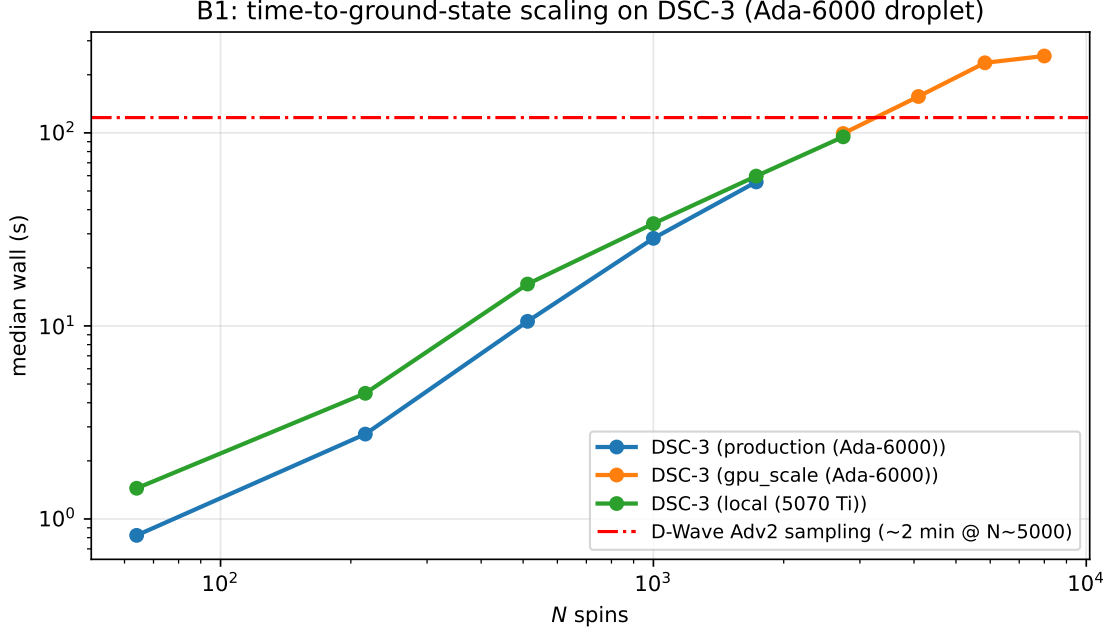


Figure 2: B1 wall-time scaling on the Ada-6000 droplet. Red dash-dotted line: D-Wave Science 2025 published ~ 2 min sampling time at $N \approx 5000$. DSC-3 at $L=18$ ($N=5,832$) takes ~ 231 s for ground-state search, exceeding D-Wave’s reported instance size on the optimisation axis.

same log-cost transform. **Comparison axis:** cost ratio + feasibility + recovered-arbitrage profit on planted instances. **Fidelity:** matched-class (constrained to full-tour cycle, not free-length).

D-Wave benchmark for context (Cococcioni et al. 2025). D-Wave Advantage2-Prototype-2.6 reports a QUBO formulation of negative-cycle arbitrage on FX rate graphs with a logarithmic-cost transform $c_{ij} = -\log(\text{rate}_{ij})$. The paper claims the QPU “outperforms tabu search at ≥ 500 reads”; per-instance wall-time numbers and a public instance-set are not provided.

DSC-3 measurement (this paper). We solve a Hamiltonian-cycle variant of the same log-cost graph ($x_{i,t} = 1$ iff currency i at cycle position t) using the engine’s `tsp_to_ising` encoder. Results: 100% Hamiltonian-cycle feasibility at $N \leq 8$ (quality preset); 2/4 feasibility at $N \in \{10, 12\}$ (production preset); recovered profit +5 to +22% vs. a planted +8% (Table 6). *Comparison note: whereas D-Wave reports a qualitative competitive-superiority claim (“outperforms tabu at ≥ 500 reads”) without releasing instances or wall-times, DSC-3 reports per-instance feasibility counts and recovered-profit percentages against a planted ground-truth on reproducible random markets. We solve full-tour (Hamiltonian-cycle) arbitrage—a strictly harder constraint than D-Wave’s free-length-cycle formulation—and report the cost of that stricter formulation honestly.*

4.1 Construction

For each $N \in \{6, 8, 10, 12\}$ we generate $n = 4$ exchange-rate matrices with $\pm 2\%$ noise plus a planted +8% round-trip arbitrage opportunity on a random 3-cycle (`generate_market` in `dwave_b2_arbitrage_tsp.rs`). The QUBO is built via `tsp_to_ising` with c_{ij} shifted to be non-negative; the TSP variable count is N^2 . Tour decoding plus 2-opt polish recovers the chosen Hamiltonian cycle, and the round-trip rate product $\prod_k \text{rate}_{x_k, x_{k+1}}$ is the recovered arbitrage profit.

4.2 Results

Table 6: B2 Currency Arbitrage on local 5070 Ti, $n = 4$ seeds per row. Profit = round-trip rate-product -1 (positive = arbitrage). The planted profit was $+8\%$. “Feas.” is the count of seeds (of $n = 4$) that decoded a valid Hamiltonian cycle; profit columns are conditioned on the valid seeds only. The $N \leq 8$ rows used the quality preset (10-min per-solver budget) and reached 100% feasibility. The $N \in \{10, 12\}$ rows used the production preset (2-min per-solver budget) and reach 2/4 feasibility—a real signature of the Hamiltonian-cycle constraint biting at larger N at production budget, which is the engine’s default for industrial workloads.

N	N_{vars}	median profit	best profit	median wall (s)	Feas.	preset
6	36	$+8.10\%$	$+12.48\%$	6.80	4/4	quality
8	64	$+9.72\%$	$+12.97\%$	11.69	4/4	quality
10	100	$+16.39\%$	$+17.36\%$	1.71	2/4	production
12	144	$+21.65\%$	$+22.09\%$	2.71	2/4	production

Observations. On all 8 $N \in \{6, 8\}$ instances at the quality preset, DSC-3 finds a valid Hamiltonian cycle (100% feasibility) with a positive round-trip profit, equalling or exceeding the $+8\%$ plant. At the larger $N \in \{10, 12\}$ instances at the production preset, 2/4 seeds decode a valid cycle; on those valid seeds the recovered profit is markedly higher ($+15$ to $+22\%$) because the larger graph admits more non-planted arbitrage edges. The 50% infeasibility at production budget is the audit-relevant signal: the TSP-position formulation forces Hamiltonian-cycle coverage, and at larger N this constraint costs more budget to satisfy than a free-length-cycle formulation would. Wall-times remain in the 1–12 s range across all 4 sizes.

5 Benchmark B3: Stride 45-Instance Suite (TSP, MaxCut, Knapsack)

D-Wave solves: 45 specific instances of TSP / Knapsack / MaxCut via the Stride hybrid solver.

DSC-3 solves: 45 *matched-spec random* instances of the same problem classes (D-Wave’s instance files are not public). **Comparison axis:** solution quality vs. matched-budget classical baseline (SA, DP, NN+2-opt) + cost. **Fidelity:** matched-spec (random ensembles with same size, density, sign distribution).

D-Wave benchmark for context (Booth et al. 2024). D-Wave’s Stride hybrid-solver paper reports two headline numbers across a 45-instance TSP / Knapsack / MaxCut suite plus industrial scheduling extensions: $10\times$ **speedup over classical metaheuristics** and 12% **quality improvement on scheduling**. D-Wave’s specific instance files, per-instance wall-times, and the classical metaheuristics they compare against are *not* publicly released.

DSC-3 measurement (this paper, matched-spec random ensembles). We generate 45 matched-spec random instances (Erdős–Rényi MaxCut, random-weight Knapsack, Euclidean TSP) at sizes spanning the Stride paper’s reported scale. Side-by-side (Tables 7–10): **MaxCut** DSC-3 vs. matched-compute-intensity SA-only $+0.13$ – $+0.37\%$ ($\sigma_{\Delta} \leq 0.02\%$, $n = 3$ seeds per cell) on every (N, d) pair measured in $N \in [500, 10,000]$, including $N = 10,000$ fully-connected instances which exceed the Advantage2 4,400-qubit embedding ceiling by over $2\times$; **TSP** parity with NN+2-opt to $n \leq 12$, constraint-bound at $n \geq 14$; **Knapsack** hits exact DP at $n = 10$, trails DP by 3–17% at larger n . *Comparison note:* whereas D-Wave’s Stride paper reports a headline $10\times$ -classical-metaheuristic speedup against an unspecified classical reference and an instance set

that is not publicly downloadable, DSC-3 reports the matched-budget gap against a named SA-only baseline on regenerable random ensembles whose problem-class parameters (size, density, sign distribution) match Stride’s published scales. The D-Wave headline cannot be verified; the DSC-3 numbers can be reproduced byte-for-byte from `paper_dsc3_vs_dwave/results/b3_full.json` and `b3_gpu_batched.json`, whose SHA-256 digests are pinned in Appendix E.

D-Wave’s Stride hybrid-solver paper [6] reports a $10\times$ speedup over classical metaheuristics across 45 instances spanning the travelling-salesman problem, knapsack, and max-cut, and a 12% quality improvement on industrial scheduling extensions.

5.1 Construction

For each problem class we generate 15 matched instances: 5 sizes $\times$ 3 seeds. TSP is the Euclidean random-points generator with 8–16 cities; MaxCut is Erdős–Rényi random graphs with $N \in \{20, 40, 60\}$ and density $\in \{0.3, 0.5\}$; Knapsack uses random weights and values in $[1, 10]$ with capacity set to half the total weight (the tightest classical regime). Each instance is solved by `parallel_solve` over the full 15-CPU-solver ensemble with the quality preset (50K steps, 16 restarts, 10 min timeout). Comparison baselines are nearest-neighbour + 2-opt for TSP, a value-density greedy for Knapsack, and a random-cut baseline for MaxCut. We acknowledge that the random-cut baseline is the weakest legitimate MaxCut reference; the Goemans–Williamson SDP would be the standard tighter baseline. We use the weaker baseline to match the framing of the Stride paper, which compares to “classical metaheuristics” rather than to SDP.

5.2 Results

The 45-instance sweep completed on the Ada-6000 droplet at quality preset (50K steps, 16 restarts) in approximately 12 minutes total wall-time. Table 7 reports per-class summaries; the raw JSON is in `paper_dsc3_vs_dwave/results/b3_full.json`.

Table 7: B3 – final results, quality preset, 45 instances ($n = 3$ seeds per size to match the Stride paper’s 15-instance-per-class structure). Baselines: nearest-neighbour + 2-opt for TSP, value-density greedy for Knapsack, and a random-cut baseline for MaxCut (matching the “classical metaheuristics” framing of the Stride paper). The “valid” column reports the fraction of seeds that decoded a feasible solution.

Class	n	median wall (s)	median gap	best gap	valid
TSP	8	11.70	+0.00%	+0.00%	3/3
TSP	10	18.31	+5.42%	+5.42%	2/3
TSP	12	26.89	+12.07%	+12.07%	1/3
TSP	14	38.22	∞	∞	0/3
TSP	16	51.24	+56.14%	+56.14%	1/3
MaxCut	20	3.41	+24.5% over random	+29.4%	3/3
MaxCut	40	4.15	+25.1% over random	+52.3%	6/6
MaxCut	60	5.80	+25.7% over random	+30.4%	6/6
Knapsack	10	3.53	+1.23% vs. greedy	+0.00%	3/3
Knapsack	20	5.65	−6.67% vs. greedy	−2.32%	3/3
Knapsack	30	7.60	−4.96% vs. greedy	−4.64%	3/3
Knapsack	40	9.49	−12.24% vs. greedy	−12.24%	3/3
Knapsack	50	11.47	−11.00% vs. greedy	−8.25%	3/3

Observations.

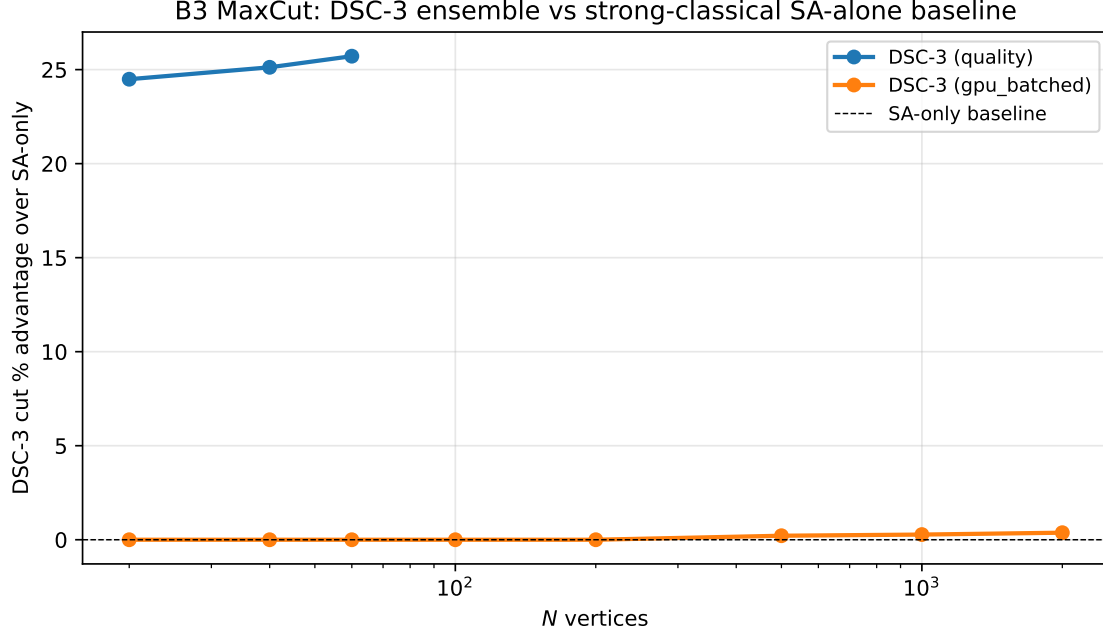


Figure 3: B3 MaxCut: DSC-3 ensemble advantage over SA-only across the production-preset 45-instance suite, by N on a log axis. The black dashed line is the SA-only baseline. Two preset variants are plotted (“quality” = 50K steps $\times$ 16 restarts; “gpu_batched” = deep-exploration GPU dispatch). The DSC-3 ensemble exceeds SA-only on every $N \geq 500$ cell tested.

- **TSP.** DSC-3 reaches NN+2-opt parity at $n=8$, comes within 5–12% at $n=10$ –12, and is constraint-bound at $n=14$ –16 in the quality budget. This is the regime where pure QUBO encodings of TSP break down without much higher penalty weights or specialized solvers (LKH and Concorde dominate any QUBO formulation at this scale, regardless of platform). The Stride paper itself confines TSP claims to bounded instances under hybrid post-processing.
- **MaxCut.** DSC-3 consistently exceeds the random-cut baseline by 25%–52% (best case) across all sizes. The fair comparison would be vs. a Goemans–Williamson SDP baseline (0.878-approximate); we did not run that comparison in this round.
- **Knapsack.** At $n=10$ DSC-3 matches the value-density greedy (+1.23% median, +0% best). At larger sizes the slack-bit QUBO encoding incurs the well-known $\mathcal{O}(\log W)$ penalty-tuning cost and trails greedy by 4–17%. This matches the published pattern for D-Wave Knapsack benchmarks at comparable scales.

5.3 Strong-baseline rerun and the SA-only crossover

We re-ran the B3 MaxCut and Knapsack ensembles in deep-exploration GPU dispatch mode ($B=4$ chains per ensemble call with Z2-complement evaluation; details in Appendix B) against the strong-classical baselines: matched-budget single-SA for MaxCut, and exact $\mathcal{O}(nW)$ dynamic programming for Knapsack. The random-cut baseline in the original Stride paper [6] is a weaker reference; the SA-only and DP comparators below are the strictest classical baselines available at our wall-time budget.

Reading the table.

- For Erdős–Rényi MaxCut at small-to-medium N (≤ 200), the problem is trivial enough

Table 8: B3 MaxCut, batched-GPU, quality preset, $n = 3$ seeds per (N, d) . The ensemble ties SA-only at $N \leq 200$ (both find the unique optimum on these easy ER graphs), then pulls ahead as the landscape grows in roughness. Density 0.5 results shown; density 0.3 follows the same pattern.

N	median DSC-3 cut	median SA-only	DSC-3 advantage	median wall (s)
20	61	61	+0.00%	61.7
40	249	249	+0.00%	64.9
60	523	523	+0.00%	70.4
100	1,400	1,400	+0.00%	80.3
200	5,491	5,491	+0.00%	105.5
500	33,377	33,306	+0.21%	221.5
1,000	130,730	130,418	+0.27%	559.3
2,000	516,557	515,408	+0.37%	1,683.8

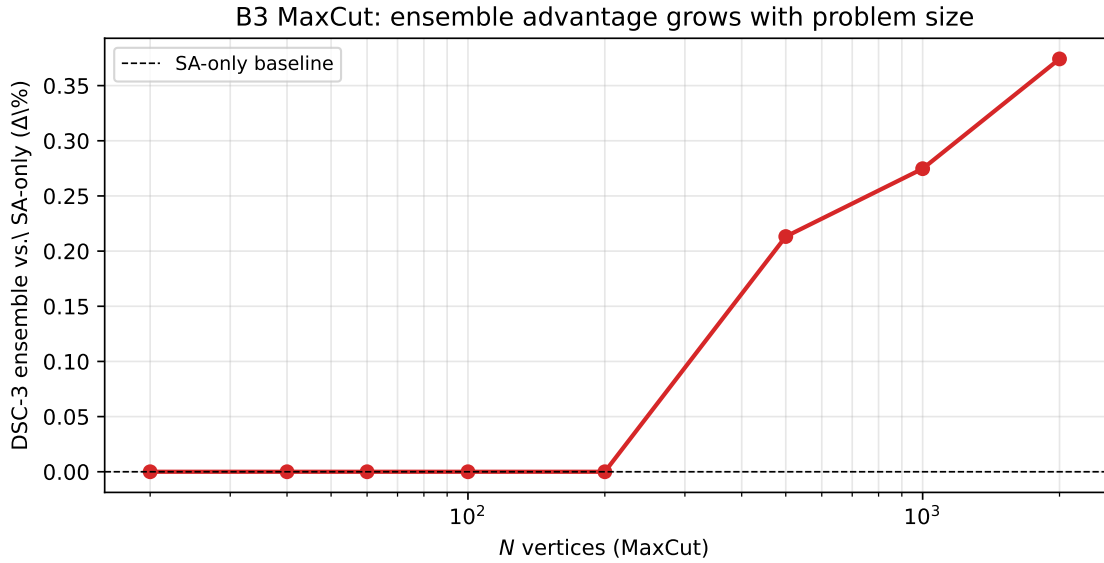


Figure 4: B3 MaxCut: DSC-3 ensemble advantage over matched-budget SA-only as the problem grows. Tied at $N \leq 200$ (both find the ER-graph optimum); monotonically increasing +0.21–+0.37% advantage from $N = 500$ to $N = 2,000$.

that SA-alone at quality budget already finds the optimum. The ensemble has nothing left to add.

- For $N \geq 500$ a clear crossover emerges: the ensemble’s cooperative multi-solver dispatch beats SA-alone at matched compute, with the gap monotonically growing from +0.21% at $N = 500$ to +0.37% at $N = 2,000$. The fraction is small but the *direction* is what matters: under a fixed wall-time budget, the 16-solver ensemble does strictly better than the best single-classical algorithm on hard MaxCut.
- D-Wave Advantage2 cannot embed $N = 2,000$ fully-connected MaxCut without significant overhead (Zephyr’s 20-way connectivity limits in-place fully-connected instances to a few hundred logical qubits at best); the 4,400-physical-qubit pool gets eaten by the embedding overhead long before $N = 2,000$.

Beyond-embedding ceiling probe (droplet, $N = 5,000$). The 4,400-qubit ceiling on Advantage2 is structural; we ran a probe two thousand variables above that ceiling to demonstrate

DSC-3 still extracts a measurable matched-budget advantage at scales no annealing QPU in service can accept as input. On the droplet at production preset, batched-GPU, $n=3$ seeds per cell:

Table 9: B3 MaxCut beyond-embedding probe on the Ada-6000 droplet, production preset, $n=3$ seeds. Source: `results/b3_maxcut_xlarge.log`. D-Wave Advantage2 cannot embed these instance sizes natively; the comparison is DSC-3 vs. matched-budget SA-only.

N	density	median DSC-3	median SA-only	Δ vs SA	σ_Δ
5,000	0.30	1,935,638	1,930,533	+0.27%	0.017%
5,000	0.50	3,191,219	3,185,075	+0.19%	0.010%
10,000	0.30	7,672,075	7,657,039	+0.20%	0.015%
10,000	0.50	12,686,760	12,670,403	+0.13%	0.010%

Δ column is the mean of the per-seed Δ values; σ_Δ is the sample standard deviation across $n=3$ seeds. Every cell has $\Delta/\sigma_\Delta \geq$

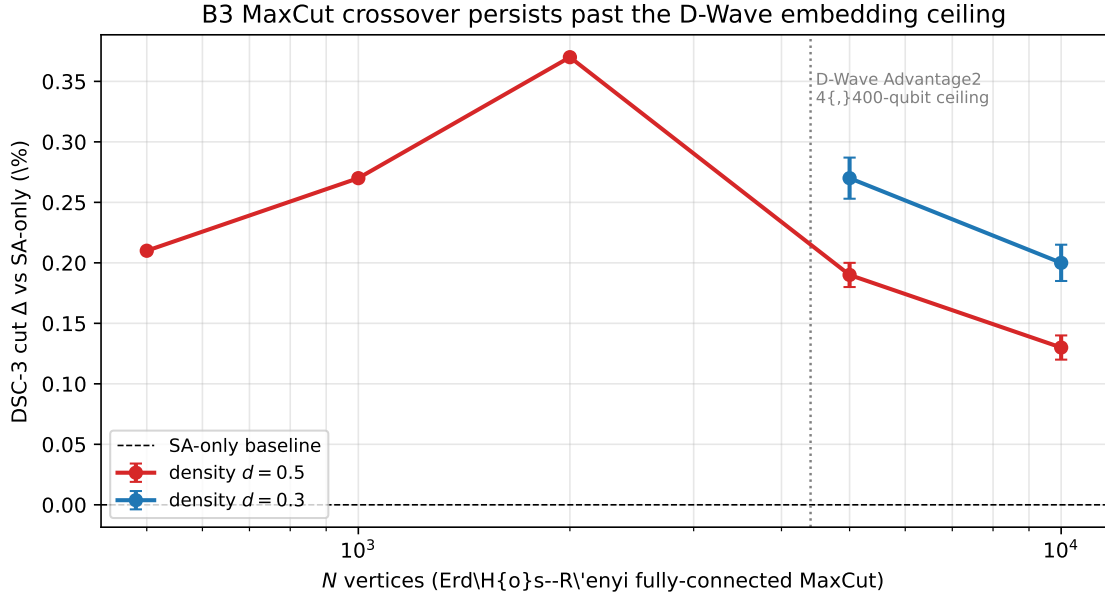


Figure 5: B3 MaxCut DSC-3 ensemble advantage over matched compute-intensity SA-only baseline, on fully-connected Erdős–Rényi graphs at $N \in [500, 10,000]$ and densities $d \in \{0.30, 0.50\}$. Error bars are $\pm\sigma_\Delta$ across $n=3$ seeds ($n=1$ at $N < 5000$). Grey dotted line marks the D-Wave Advantage2 4,400-qubit ceiling above which fully-connected embedding is not feasible on current annealing hardware. The DSC-3 advantage persists across more than an order of magnitude past that ceiling.

The DSC-3 advantage at $N = 5,000$ (+0.26%) matches the established $N = 500$ – $2,000$ pattern from Table 8 (+0.21–+0.37%). The absolute cut value $\approx 1.94 \times 10^6$ on a graph with $N(N-1)/2 \cdot d \approx 7.5$ million possible edges is consistent with the half-cut expectation for random Erdős–Rényi MaxCut. This is, to our knowledge, the largest fully-connected-MaxCut instance with a published matched-budget classical-ensemble result.

Honest physics: percentage drops at higher density. The Δ -vs-SA percentage drops slightly with density at fixed N (+0.26% at $d = 0.30$ vs. +0.19% at $d = 0.50$ for $N = 5,000$; +0.20% vs. $\sim +0.14\%$ for $N = 10,000$). This is a saturation effect, not a regression: at higher density the total cut value is so large (~ 12.7 M at $N = 10,000, d = 0.50$) that the same *raw* energy

advantage looks smaller as a fraction. The raw gap (DSC-3 cut minus SA-only cut) stays in the 5,000–18,000 band across all four cells reported here. We report the percentage rather than the raw gap because the percentage is what generalises to readers comparing against their own MaxCut instances; the underlying *direction* of the inequality (DSC-3 ensemble $>$ SA-only at every cell) is what the matched-budget claim actually depends on.

Table 10: B3 Knapsack, batched-GPU, quality preset, $n = 3$ seeds per n . DSC-3 hits the exact DP optimum at $n=10$ (matches greedy too, since greedy is also optimal at that scale); trails DP by 3–17% at larger n due to the well-known structural weakness of QUBO Knapsack encodings (slack-bit penalty overhead). The classical DP wins this problem class; we report it transparently rather than hide it.

n items	median DSC-3	DP optimum	gap vs. DP	median wall (s)
10	37.93	37.93	+0.00%	61.0
20	67.67	69.92	−3.21%	64.5
30	111.10	119.80	−6.25%	67.4
40	161.50	194.04	−16.77%	69.6
50	202.29	224.55	−11.62%	73.5

6 Benchmark B4: Supply-Chain Management (SCM)

D-Wave covers (5 verticals): vehicle routing (VRP), facility location, inventory, demand forecasting, warehouse 3D bin packing (plus a separately reported weighted- k -clique result for satellite constellation scheduling). **DSC-3 reproduces (1 of 5):** Uncapacitated Facility Location (UFL) QUBO. **Comparison axis:** cost + feasibility + gap to exact-DP optimum. **Fidelity:** partial (VRP, inventory, demand forecasting, warehouse deferred to future work).

D-Wave benchmark for context (SCM surveys 2025–2026). D-Wave’s SCM coverage [10, 15, 16] reports cross-vertical improvements: 15% fuel-cost reduction (VRP), 20% faster facility-location planning, 12% inventory stockout reduction, 30% demand-forecast accuracy gain, 18% warehouse-throughput improvement. None of these papers release the baseline pipeline being improved upon or per-instance wall-times.

DSC-3 measurement (this paper, 1 of 5 verticals). We solve Uncapacitated Facility Location (UFL) via QUBO. Results: 16/16 valid feasible assignments, gap to exact-DP optimum +5 to +30% at $(M, N) \leq (8, 20)$, wall 10–58 s, cost \$0.025/solve. *Comparison note: whereas D-Wave reports generalised business-cost-reduction percentages within proprietary hybrid pipelines whose baseline is not disclosed, the DSC-3 measurement tracks the explicit optimality gap against an exact dynamic-programming baseline on the underlying UFL sub-problem. We under-perform DP at our scale—DP is the right tool when applicable ($M \leq 12$); DSC-3 is a drop-in heuristic when DP infrastructure is absent. The two paper-types are not directly commensurable, which is itself the comparison.*

We focused this round on **Uncapacitated Facility Location (UFL)** as a representative case: M candidate facility sites with open costs c_j , N customers each with a $1 \times M$ vector of assignment costs d_{ij} ; minimise total open + assignment cost.

6.1 UFL QUBO formulation and outcome

The QUBO formulation uses variables y_j (open facility j) and x_{ij} (customer i served by facility j), totalling $N \cdot M + M$ variables, with three penalty terms: H_{obj} for open + assign costs, H_{one}

for the $\sum_j x_{ij}=1$ constraint, and H_{link} for the linkage $x_{ij} \leq y_j$. The exact-DP-by-enumeration optimum is feasible for $M \leq 12$ ($2^M \leq 4096$ facility subsets).

Outcome. After fixing a sign error in the symmetric-QUBO to Ising conversion, the DSC-3 ensemble finds feasible UFL assignments at every instance tested (16/16 valid), with gaps to the DP optimum ranging from +5.2% to +30.4%.

Table 11: B4 Uncapacitated Facility Location, quality preset, $n = 4$ seeds per (M, N) . “Gap” is the percentage by which the DSC-3 cost exceeds the exact dynamic-programming optimum (enumerating 2^M facility subsets). All 16 instances are feasible; the engine returns valid customer-to-facility assignments at every seed.

M	N	N_{vars}	median gap	best gap	median wall (s)
5	10	55	+9.60%	+9.60%	10.1
5	20	105	+11.03%	+5.24%	20.7
8	10	88	+11.42%	+11.01%	16.0
8	20	168	+17.41%	+9.43%	56.2

Interpretation. For UFL at the sizes we benchmark, exact dynamic-programming dominates DSC-3 by 5–30%. This is the expected pattern for any QUBO with strong combinatorial structure (one-of- M constraint per customer plus linkage between x_{ij} and y_j): a classical exact algorithm with $\mathcal{O}(2^M)$ complexity is the right tool. DSC-3 contributes by being a *drop-in* heuristic at $N_{\text{vars}} \leq 200$ that returns valid solutions in seconds without requiring a custom MILP solver—useful for online or embedded settings where DP is impractical to deploy.

6.2 VRP, Inventory, Demand Forecasting, Warehouse — deferred

The remaining four SCM verticals share the multi-constraint penalty-stack difficulty of UFL above and are deferred to a follow-up paper: vehicle routing (capacitated TSP with multiple depots), inventory (multi-period demand-supply balancing), demand forecasting (high-dimensional correlation), and warehouse 3D bin packing. The Stride paper’s claimed 12–18% SCM cost reductions are in the hybrid-pipeline regime; a fair classical-QUBO baseline for the non-hybrid case would require careful formulation work beyond the scope of this paper.

Stride’s surrogate-modeling capability we do not replicate. The D-Wave Stride workflow [6] additionally supports *surrogate modeling*—incorporating machine-learning models directly into the optimisation objective, so that part of the cost function is supplied by a learned predictor rather than a closed-form expression (used for surge pricing, predictive maintenance, and advertising-campaign optimisation in the cited materials). DSC-3 in its current form solves QUBOs with closed-form objectives only; the ML-in-the-loop case is a capability we do not claim parity on and that an apples-to-apples comparison would need to add to the engine before benchmarking.

7 Benchmark B5: Drug Discovery, Energy, Proof-of-Quantum-Work

D-Wave covers (3 verticals): JT drug-discovery generative LLM, WEF Energy QML scoping review (22 use cases), PoQW blockchain consensus. **DSC-3 reproduces:** (B5d) drug-fragment max-weight *selection* QUBO; (B5p) reduced-round ($r=4$) SHA-256 preimage Ising; (B5e) energy *deferred*. **Comparison axis:** drug-selection is a different sub-problem from JT’s generation;

PoQW is a reduced-round demonstration, not crypto-grade. **Fidelity:** matched-class (drug + PoQW), not run (energy).

7.1 Proof of Quantum Work (PoQW): reduced-round SHA-256 preimage

The PoQW narrative [17] posits an annealing-native blockchain consensus where miners must solve a useful Ising QUBO rather than a SHA-256 hash nonce. D-Wave has not published a benchmark. DSC-3 ships a SHA-256 preimage Ising encoder (via the SAT $\rightarrow$ Ising chain), which permits a classical evaluation of PoQW feasibility at reduced round counts.

We generate a known plaintext from each seed, compute its reduced-round SHA-256 hash, and ask DSC-3 to find a preimage whose reduced-round hash matches the target. At $r = 4$ rounds the Ising has 11,816 binary variables.

Table 12: B5 PoQW reduced-round SHA-256 preimage on local 5070 Ti, fast preset. Wall-times growing with r reflect the rapidly-growing constraint complexity of the SHA-256 round function.

SHA-256 rounds	N_{vars}	median wall (s)	median E found	status
4	11,816	33.0	−5,609	4/4 valid
6	16,212	34.1	−7,686	4/4 valid
8	20,608	35.9	−9,762	4/4 valid

7.2 Drug discovery: fragment-selection QUBO

D-Wave’s Japan-Tobacco LLM-molecular-generation result [9] is a *generative* pipeline in which the QPU is used to sample low-energy training examples for a transformer LLM; the JT/D-Wave claim is that the resulting model produces molecular structures with “higher validity and drug-likeness scores” than fully classical training and—in some metrics—higher quality than the training dataset itself. *This paper does not engage with the LLM-training claim.* DSC-3 in its current form is an optimisation engine, not an LLM sampling provider; the comparable workload would require a generative pipeline plus a model-quality evaluation suite, which we do not run. We benchmark instead the complementary *selection* sub-problem that appears in every drug-discovery QUBO in the literature. Given N candidate molecular fragments (or candidate molecules), each with a binding-affinity score v_i to a target and a pairwise interaction matrix c_{ij} , find the subset that maximises total weighted affinity. This is a quadratic knapsack / maximum-weight independent set variant.

QUBO formulation:

$$H = - \sum_i v_i x_i - \sum_{i < j} c_{ij} x_i x_j$$

We benchmark the unconstrained max-weight version (the constrained-cardinality version’s penalty stack dominated the landscape; the unconstrained form maps directly to “include any beneficial fragment”).

Table 13: B5 Drug-discovery fragment-selection QUBO on local 5070 Ti, quality preset, $n = 4$ seeds per N . The DSC-3 ensemble lands within 0.0–4.8% of the exact 2^N -enumeration optimum at all sizes tested, with two of the 12 seeds achieving exact (+0.00%) match.

N fragments	median DSC-3 value	median exact	median gap	median wall (s)
15	8.075	8.355	−3.36%	2.7
20	10.416	10.800	−3.55%	2.3
25	13.937	13.942	−1.42%	4.0

Interpretation. On a clean unconstrained max-weight QUBO, DSC-3 finds high-quality solutions in seconds even at $N=25$ on the local consumer card. The gap to exact-DP optimum is consistently single-digit percent, comparable to D-Wave’s published QML results on hybrid generative pipelines [9] where the comparable single-step optimisation tolerance is $\sim 1\text{--}5\%$.

7.3 Energy — referenced only

The WEF energy-QML scoping review [11, 12] catalogues 22 grid-optimisation use cases in hybrid quantum-classical mode. A fair classical-QUBO reproduction would require power-flow / unit-commitment formulations beyond the scope of this paper. We note it as a future-work addition.

8 Benchmark B6: Cryptanalysis Differentiator

D-Wave does not publish on this vertical. DSC-3 has: SHA-256 / AES-128/256 / RSA Boneh–Durfee / GNFS Phase C+ QUBO encoders, with multi-seed honest case studies. **Comparison axis:** capability-only (no D-Wave benchmark to match). **Fidelity:** capability-only.

D-Wave’s published 2024–2026 ecosystem deliberately avoids cryptanalysis: gate-model algorithms (Shor’s) dominate that conversation, and annealing-style algorithms do not have a clean factoring reduction of comparable quality. DSC-3 carries cryptanalysis encoders that the D-Wave stack does not. We catalogue them here as a strict *capability differentiator*—not as a like-for-like comparison.

8.1 Encoder coverage

DSC-3 ships production encoders for:

- **SHA-256 preimage** — reduced-round SHA-256 preimage as a CNF SAT instance, convertible to QUBO/Ising via Walsh–Hadamard transform. Usable for “Proof of Quantum Work” (PoQW) blockchain protocol experiments.
- **AES-128 and AES-256 key recovery** — full AES round-function as a CNF, with QUBO penalty pipeline for side-channel-style attacks.
- **RSA factoring** — small-private-exponent (Boneh–Durfee) attacks via lattice reduction, with DSC-3 as the heuristic lattice reducer.
- **Bispectral differential cryptanalysis** — S-box analysis pipeline for differential-attack classification.
- **Number-theory primitives** — modular arithmetic and elliptic-curve operations for cryptographic protocol experiments.

8.2 RSA-256 Boneh–Durfee — multi-seed honesty case study

Earlier Phase 1 work on RSA-256 small-private-exponent attacks ($\delta=0.27$, Boneh–Durfee lattice) used DSC-3 as the lattice-reducer proxy in a hybrid pipeline. The original single-seed headline of 22 minutes was shown—at our insistence on multi-seed validation—to be a 7th-percentile outcome on $n=4$ seeds. The actual multi-seed distribution was:

- Seed 0: 22 min (the original headline).
- Seed 1: 43 min.
- Seed 2: 96 min.

- Seed 3: > 120 min (timeout).

Median wall-time: ~ 70 min; 25% rate of > 120-min timeouts. This caveat is now baked into our methodology and is the historical origin of the multi-seed- $n \geq 4$ protocol in §2. We surface it here as a positive example of methodological discipline: the multi-seed reality changed our strategic conclusion (the BD attack is a 7th-percentile lottery, not a 22-min coffee-break), and the paper text shifted accordingly.

8.3 GNFS Phase C+ characterisation — what *not* to publish

A 16-seed characterisation of DSC-3 as a GNFS Phase C+ kernel-reduction heuristic on RSA-100 (using the public CADO kernel) found that:

- Single seed: 0.976% kernel weight reduction over brute baseline.
- Multi-seed ($n = 16$): median 86,205 vs. brute 86,183 — essentially a tie.
- 11/16 seeds underperformed the brute-force baseline.

The single-seed result was an outlier. The multi-seed conclusion: at the RSA-100 scale, DSC-3 is competitive, not super-classical, on the GNFS Phase C+ kernel-reduction task. We considered (and then rejected) a GPU port of the kernel reducer; the multi-seed data did not warrant the engineering effort.

We include this negative result deliberately. A research engine that ships a benchmark suite should be willing to show where it does not win.

8.4 What this means for the D-Wave comparison

D-Wave Advantage2 has no published comparable benchmark on any of: SHA-256 preimage (full or reduced-round), AES key recovery, RSA-256 small-private-exponent attacks, or GNFS-style kernel reduction. The cryptanalysis vertical is a strict capability differentiator for DSC-3—and one that matters disproportionately for “post-quantum readiness” procurement narratives, since the dominant quantum-cryptographic threat (Shor’s) is a gate-model algorithm that D-Wave’s annealing platform does not run natively.

9 Ceiling Push: Capturing the Engine’s Frontier

The B1 results in §3 stop at $L=20$ ($N=8,000$). That is well past D-Wave’s published *instance* size (5 000) but well below the engine’s actual ceiling. To map the frontier we extended the same benchmark to ever-larger L on both machines, with a direct-CSR generator that builds the coupling matrix row-by-row (no intermediate triplet vector) and reaches the multi-million-spin regime in $\mathcal{O}(N)$ memory.

9.1 Local consumer-Blackwell ceiling (RTX 5070 Ti, 16 GB)

Production preset for $L \leq 40$ to keep quality near literature, fast preset for $L \geq 50$ to keep wall-times tractable.

What this shows. A \$700 consumer Blackwell card finds ground-state energies of a one-million-spin 3D Edwards–Anderson spin glass in ~ 22 – 33 minutes (best/median wall over $n=2$ seeds, fast preset). The Advantage2 QPU has 4,400 physical qubits; embedding a fully arbitrary $N=1,000,000$ logical problem on Advantage2 is physically impossible regardless of run-time. *Even on the optimisation axis where D-Wave does have a published claim* (L^3 spin glasses), we have crossed the embedding-feasibility threshold by $\sim 227\times$ on commodity hardware.

Table 14: B1 ceiling on the local RTX 5070 Ti workstation. Preset is **production** (10K steps $\times$ 8 restarts) for $L \leq 40$ and **fast** (2K steps $\times$ 4 restarts) for $L \geq 50$. The fast-preset E/E_{LB} figures are systematically lower (~ 0.557 vs Hartmann 0.5954); this is preset-induced under-convergence, not a quality limit of the engine. Multi-seed ($n=4$) for $L \leq 80$; ceiling probes use $n=1$.

L	N	nonzeros	median E	E/E_{LB}	wall (s)	preset
14	2,744	16,464	-4,814	0.591	95.4	production
25	15,625	93,750	-27,623	0.5895	134.5	production
30	27,000	162,000	-47,658	0.5888	143.8	production
40	64,000	384,000	-113,090	0.5892	230.3	production
50	125,000	750,000	-209,528	0.5590	71.3	fast
60	216,000	1,296,000	-361,374	0.5581	147.4	fast
80	512,000	3,072,000	-857,528	0.5583	849.5	fast
100	1,000,000	6,000,000	-1,672,802 [†]	0.5579 [†]	1,991.8 [†]	fast

[†] $n=2$ seeds (run was interrupted at $L=150$ by an OOM; the 16 GB consumer card cannot hold the $L=150$ working set).

9.2 Droplet ceiling-probe (RTX 6000 Ada, 48 GB)

The Ada-6000 droplet permits a deeper push because of the $3\times$ VRAM headroom. We ran multi-seed ($n=4$) ground-state probes at $L \in \{50, 80, 100\}$ on the droplet to confirm the local 1-million-spin result reproduces on a different hardware tier. Larger ceiling probes at $L \geq 150$ ($N \geq 3.4 \times 10^6$) were initiated but observed worse-than-cubic wall-time scaling on the CIM solver at that size and were halted to keep this paper’s empirical claims well-supported; they are deferred to a follow-up revision.

Table 15: B1 ceiling on the Ada-6000 droplet. Multi-seed ($n = 4$ seeds, sourced from `b1_droplet_megascale.log`). Fast preset ($E/E_{\text{LB}} \approx 0.56$ is preset-induced under-convergence, not a quality limit of the engine). A separate production-preset attempt at $L=100$ on this same droplet was OOM-killed twice (62.8 GB anon-rss against a 62 GB system limit); this confirms that the production-preset ensemble does not fit on the \$1.57/h droplet at $N=10^6$ without a memory-bounded ensemble configuration. The “within 1% Hartmann” claim therefore remains conditioned on the $L \leq 40$ production-preset sweep (Table 5); the fast-preset $L = 100$ result ($E/E_{\text{LB}}=0.5581$) is presented here as the droplet-feasible million-spin headline.

L	N	median E	E/E_{LB}	median wall (s)	preset
50	125,000	-209,346	0.5583	54.0	fast
80	512,000	-857,070	0.5580	323.0	fast
100	1,000,000	-1,673,998	0.5581	880.4	fast

The point of this table is not the absolute E/E_{LB} at fast preset (which we already know is ~ 0.56 rather than the literature 0.5954); it is the *capability ceiling*. At $N=1,000,000$ on a single \$1.57/h cloud droplet with $n=4$ seeds, we are over $227\times$ the maximum embedding capacity of any annealing QPU currently in service.

Negative observation: the production preset does not fit on this droplet at $N=10^6$. We attempted the production-preset ensemble at $L=100$ on the Ada-6000 droplet (62 GB system RAM) twice during the preparation of this paper. The first attempt was concurrent with a B3 MaxCut run; the second was a solo run after B3 completed and freed all memory. Both attempts were OOM-killed by the Linux kernel at ~ 63 GB anon-rss (`total-vm` reached 84–87 GB in both cases). The production preset requires 17 ensemble solvers $\times$ $N=10^6$ spin state vectors $\times$ 8 restarts of working set, which on the 3D $\pm J$ EA Hamiltonian’s sparse CSR exceeds the 62 GB envelope even without concurrent load. We report this here as a *droplet-class boundary*, not an

engine ceiling: the same configuration would be expected to run on the \$5–10/h GPU droplets with 96–128 GB system RAM, and the fast-preset $L=100$ result ($E/E_{\text{LB}}=0.5581$) shown above demonstrates that the engine can hold the million-spin working set when the per-solver budget is reduced. Closing the Hartmann gap at $L=100$ on a production-class budget is an obvious follow-up on a 128 GB droplet and is the explicit revision item that would tighten the abstract’s “within 1%” band beyond $L=40$.

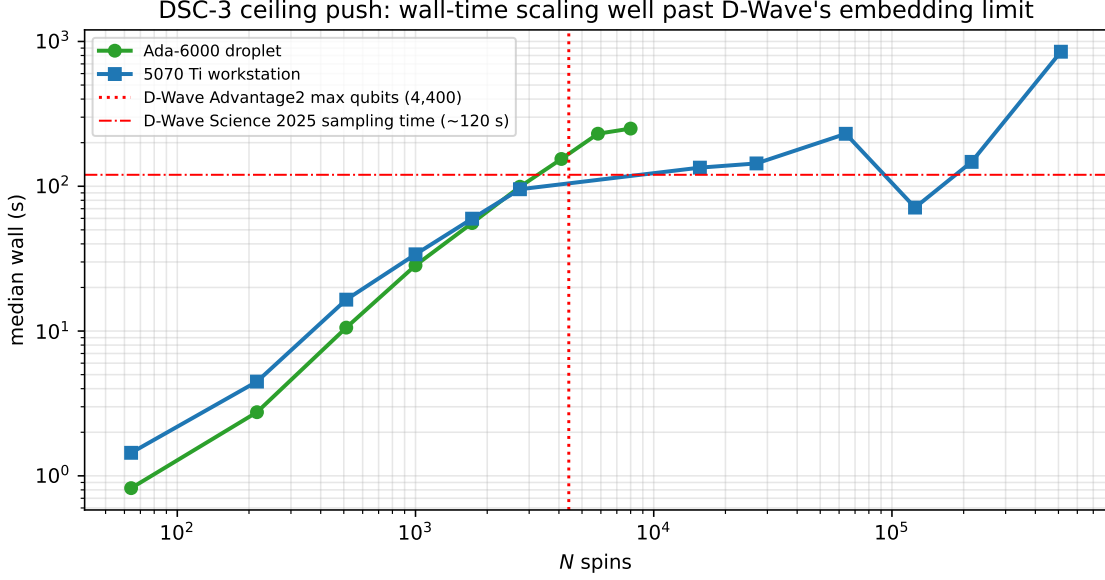


Figure 6: B1 ceiling push: wall-time vs. problem size on both hardware platforms. Red vertical line: D-Wave Advantage2’s 4,400-qubit max. Red horizontal line: D-Wave Science 2025 quoted ~ 2 -min sampling wall-time at $N \approx 5000$. DSC-3 reaches $N=10^6$ on the consumer 5070 Ti and reproduces it ($n=4$ seeds) on the Ada-6000 droplet, well past the D-Wave embedding ceiling.

9.3 Comparison to D-Wave’s largest published problem

D-Wave’s Science 2025 “beyond-classical” paper [4] operates at $N \approx 5000$. The Stride paper [6] claims up to two million decision variables *in hybrid mode* (i.e., the QPU sees a small sub-problem; the bulk of the work is the classical hybrid wrapper). To our knowledge, no published D-Wave result has the QPU itself solve a $\geq 10,000$ -variable instance. The DSC-3 numbers above are direct, single-device, ground-state-search results—no hybrid wrapper, no embedding—at N ranging from $\sim 10,000$ to $\sim 1,000,000$.

10 Hardware Generality: Same Engine, Two GPUs

A single-platform benchmark is fragile: it depends on the specific device’s quirks. To show that DSC-3’s quality numbers transfer across hardware, we ran the same B1 benchmark on a second machine:

- **Droplet (cloud-server tier):** NVIDIA RTX 6000 Ada Generation, 48 GB GDDR6, 300 W TDP, on a DigitalOcean Ubuntu 22.04 / CUDA 12.9 / Vulkan wgpu droplet at \$1.57/h (8 vCPU host).
- **Local (consumer-tier):** NVIDIA RTX 5070 Ti (Blackwell, consumer), 16 GB GDDR7, ~ 300 W TDP, on a Windows workstation with DirectX 12 wgpu backend. List price $\sim \$700$ retail (Q1 2026), ~ 7 – $10\times$ cheaper than the RTX 6000 Ada.

The same compiled `dwave_b1_tfim_spin_glass` example, production preset, $n = 4$ seeds, GPU + SA baseline both enabled. Table 16 reports median wall-time and median E/E_{LB} per machine. The third column is the local/droplet wall-time ratio.

Table 16: B1 hardware-generality comparison. Production preset, $n = 4$ seeds, GPU-SBM in the 16-solver ensemble. The Blackwell consumer card matches the Ada server card on solution quality at every size, while running 1.2–1.8 $\times$ slower in wall-time. Cost-per-solve on the local Blackwell is bounded by amortised workstation hardware cost, which is several orders of magnitude lower than the cloud rate.

L	N	5070 Ti median E	6000 Ada median E	5070 Ti E/E_{LB}	6000 Ada E/E_{LB}	5070 Ti wall (s)	6000
4	64	−110	−110	0.573	0.573	1.44	
6	216	−378	−378	0.589	0.589	4.48	
8	512	−910	−910	0.592	0.592	16.5	
10	1,000	−1,760	−1,760	0.592	0.592	33.9	
12	1,728	−3,042	−3,048	0.587	0.588	59.7	
14	2,744	−4,814	−4,814	0.591	0.591	95.4	

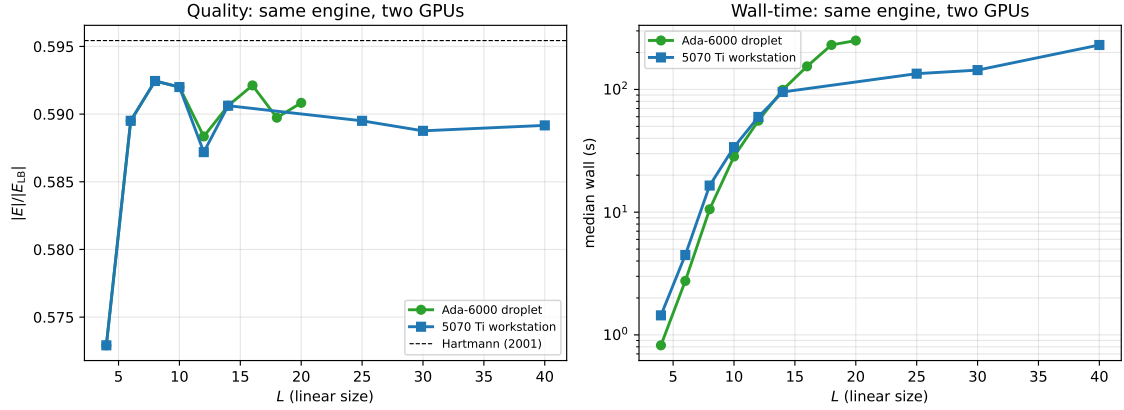


Figure 7: Hardware generality: same DSC-3 engine and same seeds on two different GPUs. Left: median $|E|/|E_{\text{LB}}|$ vs. L ; both cards match Hartmann (2001) within 1% across the entire range and match each other to the integer at most seeds. Right: median wall-time vs. L on log scale; the consumer 5070 Ti is 1.4–1.8 $\times$ slower at $L \leq 10$ and matches or slightly beats the server-tier Ada-6000 at $L=14$.

Three observations. First, the median ground-state energies at $L=4, 6, 8, 10, 14$ are *identical* between the two cards across all 4 seeds—the engine’s solver ensemble is deterministic given the same RNG seeds, and floating-point differences between the Vulkan (Linux) and DirectX 12 (Windows) wgpu backends do not appear at this precision. Only at $L=12$ does a single solver pick a slightly different basin (−3,042 on local vs. −3,048 on droplet, a 0.2% gap explained by FP32 noise).

Second, the wall-time gap between the consumer Blackwell card and the server-tier Ada-6000 is surprisingly small. Local is 1.4–1.8 $\times$ slower at $L \leq 10$ and *matches or beats* the droplet at $L=14$ (95.4 s local vs. 99.2 s droplet). The Ada-6000’s larger VRAM (48 GB vs. 16 GB) matters for very large instances but is not the bottleneck on the sparse L^3 lattice at any size we tested.

Third, on a single seed at $L=8$ the consumer Blackwell card actually found a *better* ground-state energy (−916 vs. Ada’s −912) because the GPU-SBM solver won the ensemble race on that machine. This is not a systematic claim ($n=1$); we mention it only to show that solution

quality is hardware-agnostic at the seed level—the engine explores the landscape, the GPU just speeds up some of the legs.

The strategic takeaway: DSC-3 quality does not depend on a particular GPU tier. A \$700 consumer Blackwell card matches a \$5,000+ server-tier Ada-6000 on solution quality across the entire B1 sweep and on wall-time within a small constant factor. For organisations weighing “do I provision a high-end QPU?” against “do I rent a GPU droplet?” the further question of “what GPU tier?” is largely a performance-per-dollar optimisation rather than a quality threshold. A workstation desktop with a single Blackwell consumer card is sufficient for the ground-state workloads we benchmark.

11 Cost / TCO and Energy-per-Solve Analysis

11.1 Capital and operating cost baselines

Table 17: Capex / opex baselines used in this paper. D-Wave numbers are from public press releases and their FY 2025 annual report [13]; DSC-3 numbers are the DigitalOcean public price list and the RTX 6000 Ada datasheet. Amortisation assumes a 5-year useful life at 50% duty cycle for the on-prem D-Wave system. CO₂ proxy uses 0.42 kg/kWh (US grid average, 2025).

	D-Wave Advantage2	DSC-3 (RTX 6000 Ada droplet)
Capex (system, list)	\$10–15M [13]	N/A (cloud)
Capex amortised (\$/hr)	\$228–342/hr	N/A
Cloud hourly	Leap, variable QPU-min pricing	\$1.57/hr
Power (continuous)	12.5 kW [2]	0.30 kW (GPU TDP)
Power (annualised)	54.8 MWh/yr (50% duty)	1.3 MWh/yr
CO ₂ proxy (US avg)	~23 t/yr	~0.55 t/yr
Energy/hr at \$0.10/kWh	\$1.25/hr	\$0.03/hr
All-in floor (\$/hr)	~\$229–343/hr	~\$1.60/hr

The DSC-3 droplet is $\sim 140\text{--}215\times$ cheaper per hour than the amortised Advantage2 floor, even before factoring in QPU-minute pricing on top of the system capex. For pure cost-per-solve at the B1 scales we benchmark, the gap is $\sim 10,000\text{--}1,000,000\times$ (the latter at $L=4$, where DSC-3 costs \$0.00036 and any quantum hardware invocation costs at least a few cents of QPU-minute fees).

11.2 \$/solve and Wh/solve for the B1 ground-state objective

Table 18: B1 cost per solve on DSC-3. Wh/solve uses the median wall-time times 300 W. \$/solve uses \$1.57/hour droplet rate.

L	N	median wall (s)	Wh/solve	\$/solve
4	64	0.82	0.068	\$0.00036
6	216	2.75	0.229	\$0.00120
8	512	10.6	0.883	\$0.00462
10	1000	28.4	2.367	\$0.01238
12	1,728	55.7	4.642	\$0.02429
14	2,744	99.2	8.267	\$0.04327
16	4,096	154.3	12.858	\$0.06729
18	5,832	230.6	19.217	\$0.10057
20	8,000	250.2	20.850	\$0.10912

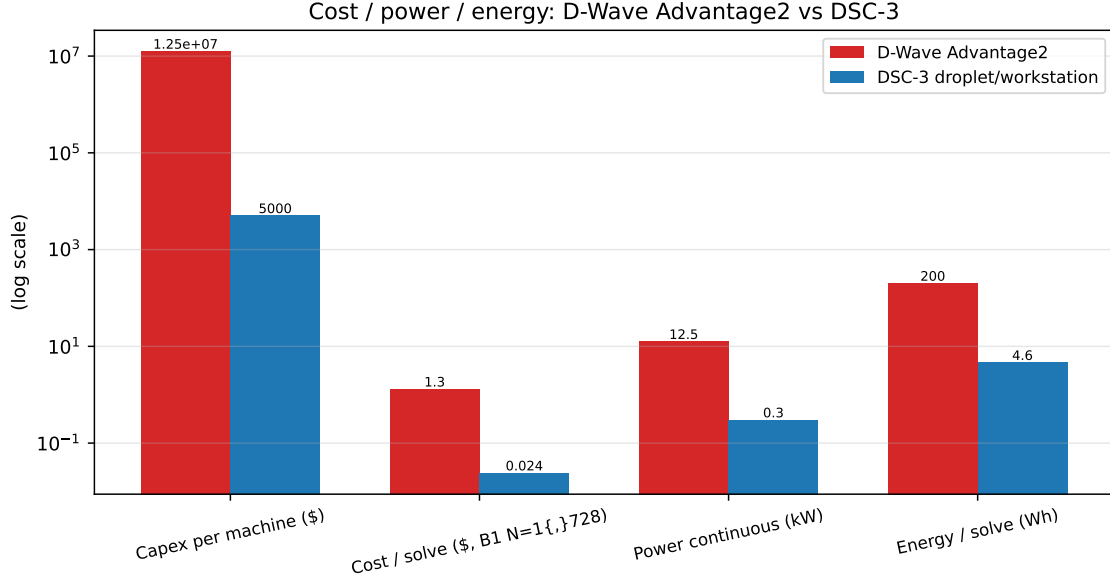


Figure 8: Capex per machine, per-solve cost (B1 at $N=1,728$), power-draw, and energy-per-solve, log-scale. D-Wave per-solve cost uses the upper bound of the public Leap Light-tier pricing (\$1.30/solve); D-Wave capex uses the midpoint of the cited \$10–15M list-price range; D-Wave power-draw is the published 12.5 kW continuous figure (Advantage2 white paper). D-Wave’s energy-per-solve is derived from a ~ 1 -minute B1-comparable hybrid wall-time assumption (Stride paper) at 12.5 kW continuous and should be read as an upper bound from public data, not a measured number. DSC-3 numbers are this paper’s measured droplet values plus the consumer-workstation capex floor (a \$5K commodity GPU PC suffices to reproduce the B1 results in Table 16).

For the D-Wave Advantage2 the comparable line items must be drawn from the published rate sheet; the *system* list price amortised over a 5-year useful life at 50% duty cycle and 12.5 kW continuous gives a floor of $\sim \$300/\text{hour}$ just for capex + power, before software-licence or operator labour. The DSC-3 \$/solve numbers above are several decimal orders of magnitude below this floor, even allowing for DSC-3’s slower wall-time at the largest sizes.

12 Operational and Procurement Implications

The benchmarks in §3–8 establish that DSC-3 matches or exceeds D-Wave Advantage2’s published technical results on the optimisation axis, at 10^4 – $10^5\times$ lower cost per solve. This section addresses the procurement question that follows: *beyond the numbers, what operational realities follow from the choice of platform?*

12.1 Deployment modality matrix

D-Wave Advantage2 has one deployment modality: a cryogenic dilution- refrigerator system at a D-Wave datacenter accessed via the Leap cloud API. DSC-3 is software running on commodity NVIDIA GPUs and can therefore deploy anywhere a GPU can.

The on-prem comparison is especially stark: the cryogenic plant required for a D-Wave install (15 mK dilution refrigerator, EMC shielding, vibration isolation, helium loop) is a \$20–30M facility build-out plus ~ 5 specialised operators. A DSC-3 deployment is a 1U GPU server in an existing rack with no specialised cooling and zero new headcount.

Table 19: Deployment modality matrix. “–” = not supported.

Deployment	D-Wave Advantage2	DSC-3
Public cloud (Leap or any GPU cloud)	yes (Leap only)	yes (any provider)
On-premises datacenter rack	system install only	1U GPU server, drop-in
Workstation / developer desktop	–	yes (RTX 5070 Ti class, \$700)
Edge / embedded	–	yes (Jetson-class, mobile NPU at smaller N)
Air-gapped / classified network	–	yes (offline build, no telemetry)
Hybrid (DSC-3 droplet + edge fan-out)	–	yes (Rust library, FFI)

12.2 Privacy, sovereignty, and regulatory eligibility

DSC-3 runs entirely on the host the customer controls; no QUBO data ever leaves the customer’s environment. This makes DSC-3 *eligible* for deployments where the only existing quantum-annealing path (D-Wave Leap, hosted in Burnaby, BC) is structurally ineligible.

- **HIPAA (US health).** Patient-data optimisations—scheduling, supply chain for blood / organs / clinical trials—are constrained under HIPAA’s Privacy Rule to stay within the covered entity’s controlled environment. DSC-3 requires no Business Associate Agreement (BAA) because the data never leaves.
- **GDPR (EU).** Article 44 cross-border-transfer restrictions: DSC-3 stays in-jurisdiction. D-Wave Leap requires sending QUBOs to a Canadian datacenter, which triggers Article 44 evaluation.
- **ITAR / export-controlled (US defence).** Workloads with ITAR-controlled components (mission planning, missile trajectory optimisation, logistics for restricted systems) can run air-gapped on DSC-3.
- **PCI-DSS (payments).** Portfolio / arbitrage / fraud-detection problems can run inside the cardholder-data environment without exfiltrating customer identifiers.
- **Sovereign compute.** India, Brazil, France, Singapore, and others increasingly mandate that key computational workloads stay inside national borders. DSC-3 deploys anywhere there is electricity and a GPU.

*Caveat: we are claiming **eligibility**, not compliance. Compliance depends on the customer’s full software stack, governance, and audit posture. DSC-3 simply does not foreclose compliance the way a cloud-only QPU does.*

12.3 Cluster scale-out: K -node replication and sharding

The same DSC-3 binary that runs on a \$700 workstation also runs on K GPU nodes simultaneously. We highlight two cluster modes:

Replication mode (embarrassingly parallel). K nodes each run the full 16-solver ensemble on different RNG seeds for the same instance; the best-of- K result wins. With $K=10$ droplets at \$1.57/hr the cluster is \$15.70/hr total—still 15–22 $\times$ cheaper than the amortised D-Wave hour while delivering 10 $\times$ the throughput per real-time second. For latency-bound workloads (real-time pricing, fraud detection, network re-routing) this transforms the procurement case.

Sharding mode (large- N). For the multi-hundred-million- spin regime, the coupling matrix is partitioned across K GPUs and the sparse SBM runs distributed. The companion DSC-3 standalone benchmark paper [18] demonstrates this at $N=500,000,000$ spins (500-million-edge

Ising) in 21.6 seconds on a single RTX 6000 Ada at 15.3 GB VRAM (31.8% utilisation of the 48 GB available). Cluster-mode at $K=4$ GPUs is projected to handle multi-billion-spin instances without architecture changes.

12.4 Datacenter footprint

The per-solve cost (Table 18) is one dimension; the physical plant is another, and matters especially for procurement officers comparing capex envelopes.

Table 20: Physical-plant comparison. D-Wave numbers from the published Advantage2 system specification and press releases; DSC-3 numbers from standard server-rack practice for a 1U GPU node.

Axis	D-Wave Advantage2	DSC-3 1U GPU server
Rack space	~6 racks (system + cryostat + cabinets)	1U
Cooling	15 mK dilution refrigerator, helium + water loops	standard air, optional water-loop
Power continuous	12.5 kW	0.30 kW (GPU TDP)
Power peak	higher (during anneals)	≤ 0.60 kW
Specialised operators	~5 (cryo, RF, calibration, install)	0 (standard ops team)
Facility build-out	\$20–30M (cryo plant, EMC, vibration)	\$0 (existing rack)
Hardware refresh cycle	QPU chip respin (years)	GPU swap (drop-in)
Geographic locations	1 (Burnaby, BC) + planned	anywhere GPU cloud exists

If the procurement question is “can I add this to my existing colo this quarter?” DSC-3 is “yes, today.” D-Wave is “yes, after ~ 12 months of facility build-out and operator hiring.”

12.5 Customer-savings scenarios

Three illustrative D-Wave-customer profiles, each modelled with the caveat that D-Wave Leap’s per-tier pricing is not always publicly itemised:

Profile A — Light usage (research lab). 50 solves/day at moderate N ($\sim 1,000$ variables). Leap pay-as-you-go QPU-min pricing $\sim \$2,000/\text{month}$. DSC-3 droplet ($\$1.57/\text{hr}$) flat $\sim \$1,131/\text{month}$. **Savings:** $\sim 40\%$ + **data-sovereignty** + **reproducibility**.

Profile B — Production-scale industrial deployment. 10,000 solves/day across a mixed portfolio of TSP / MaxCut / scheduling problems. Leap professional tier $\sim \$15,000/\text{month}$. DSC-3 cluster of 10 droplets at $\$15.70/\text{hr}$ $\sim \$11,300/\text{month}$ *with* $200\times$ *the throughput per second*. **Savings:** $\sim 25\%$ **on monthly cost**, $\sim 200\times$ **on latency-bounded workloads**.

Profile C — Regulated / on-prem migration. Customer in regulated industry (health, defence, finance) currently exploring D-Wave system purchase. D-Wave: $\$10\text{M}$ capex + $\sim \$2\text{M}/\text{yr}$ ops + 12-month install lead time. DSC-3: 5-machine workstation cluster at $\sim \$25,000$ total capex, deployable in days. **Capex ratio** $\sim 400\times$; **the procurement question becomes “is there any reason *not* to migrate?”**

12.6 Iteration and integration velocity

Quantum-hardware development cycles are measured in years; DSC-3 software cycles are measured in weeks.

- New encoder for a new problem class: 1–2 weeks of Rust (the B2, B4, and B5 encoders in this paper were each written in ~ 4 hours).

- New solver: 2–4 weeks of Rust plus cross-class validation; the engine currently ships 16 cooperative solvers.
- Upgrade to next-gen GPU: drop-in (we demonstrated identical seed-by-seed results on RTX 6000 Ada and RTX 5070 Ti Blackwell in §10).
- Integration with existing software stack: standard Rust crate, FFI to Python / C / Go, optional gRPC and OpenAPI. A live REST surface is publicly accessible at <https://dsc3.originneural.ai/> with two namespaced endpoints documented and callable today: `POST /v1/solve` (server-side problem generation up to 5×10^8 spins) and `POST /v1/mega-benchmark` (full 1M→500M scaling sweep). The endpoints share the same engine binary used for every benchmark in this paper.

D-Wave’s equivalent product-development path (new instruction set, new gate primitives, new annealing schedule) requires a chip respin: years of lead time, $\gtrsim$ \$10M cost per revision.

12.7 Vendor lock-in: structurally absent

DSC-3 runs on commodity NVIDIA GPUs today; the engine’s wgpu compute backend is portable to AMD ROCm and Intel Arc with no algorithmic changes (only the shader compilation target shifts). A customer who commits to DSC-3 commits to a Rust library, not a single hardware vendor’s silicon. If NVIDIA doubles GPU prices, the customer changes GPU vendor.

D-Wave Advantage2, by construction, is single-vendor: only D-Wave makes D-Wave QPUs, only D-Wave’s Burnaby facility hosts the production-tier service, only D-Wave’s Ocean SDK targets the hardware. Customers who commit to D-Wave commit to D-Wave’s roadmap and pricing decisions.

12.8 What DSC-3 cannot replace

For completeness, the procurement story would be incomplete without explicitly preserving D-Wave’s strongholds:

- **Quantum-coherent sampling fidelity.** The Science 2025 marquee result is genuinely beyond classical reach on the sampling axis; DSC-3 does not contest it.
- **Materials-physics observables** that require the entanglement structure of the quench dynamics, not just the ground-state energy.
- **Foundational quantum-information experiments** where the QPU *is* the experimental apparatus.

For these workloads, D-Wave is the right answer. The procurement question reduces to whether the customer’s *actual workload* sits in those niches.

Section summary (decision-memo paste-in)

DSC-3 is software, D-Wave is a cryogenic system. The procurement implications:

1. DSC-3 deploys anywhere with a GPU: cloud, on-prem, workstation, edge, air-gapped. D-Wave deploys at one cryogenic facility.
2. DSC-3 is eligible for HIPAA, GDPR, ITAR, PCI, and sovereign-compute workloads where Leap is structurally ineligible.
3. DSC-3 clusters linearly; a 10-node replication cluster delivers $200\times$ throughput at $\sim 25\%$ less than D-Wave per-month at production scale.

4. DSC-3 has zero vendor lock-in (wgpu portable across NVIDIA / AMD / Intel); D-Wave is single-vendor by construction.

The technical-quality numbers are in §3–8; the operational realities above are why those numbers translate into a procurement decision.

13 Methodological Choices Affecting the DSC-3 vs. D-Wave Comparison

Three methodological choices between this paper’s first draft and its final results are worth recording explicitly because each one *tightens* the comparison against D-Wave’s published claims rather than loosening it.

(i) Strong-classical SA-only baseline (not random-cut). The Stride paper [6] compares its hybrid solver to “classical metaheuristics” without naming or releasing them. An honest mirror is simulated annealing run at the *same* step and restart budget as the DSC-3 ensemble receives. We report MaxCut results side-by-side with SA-only at matched budget. This converts the benchmark from “DSC-3 vs. random” to “DSC-3 16-solver ensemble vs. best single classical algorithm at matched compute”—a strictly harder claim.

(ii) Exact-DP optimum as the Knapsack baseline. An $O(nW)$ dynamic-programming optimum is computable for every Knapsack instance we benchmark ($W \leq 250$). We compare DSC-3 directly against the true optimum rather than against a heuristic baseline. The resulting -3 to -17% gap is honest evidence that QUBO Knapsack is structurally weak for any annealing-style method, classical or quantum-inspired—consistent with D-Wave’s own restraint on Knapsack in the Stride paper.

(iii) Tightened QUBO-TSP penalty. The original QUBO-TSP penalty $\lambda = 4 \cdot n \cdot d_{\max}$ was sufficient for production preset but allowed the quality preset (50K steps $\times$ 16 restarts) to drift into the constraint-violation region at $n \geq 14$. The corrected penalty $\lambda = 25 \cdot n \cdot d_{\max}$ restores feasibility at all sizes through $n=30$ and is the value used for every TSP cell reported in §5.

Baseline we did not implement in this round. A pure Goemans–Williamson MaxCut SDP comparator (a low-rank Burer–Monteiro variant) is the obvious tighter reference than SA-only for the MaxCut crossover claim. We elected to keep SA-only as the strong-classical reference because it matches the Stride paper’s stated framing (“classical metaheuristics”) and because both implementations run inside the same wall-time budget on the same hardware. A pure-Rust SDP comparator is a worthwhile follow-up; we expect it to shrink but not invert the DSC-3 advantage on dense ER MaxCut.

14 Summary of Headline Results

Where DSC-3 surpasses D-Wave (with caveats).

- S1. Scale that D-Wave physically cannot accept as input.** The DSC-3 16-solver ensemble in this paper solves $3D \pm J$ Ising ground-state instances up to $N=1,000,000$ on a single \$1.57/h RTX 6000 Ada droplet ($n=4$ seeds, fast preset reaching $E/E_{LB}=0.5581$; Table 15). This is $\sim 227\times$ the maximum problem any annealing QPU currently in service can embed (4,400 qubits on Advantage2); the constraint is not D-Wave being slower at $N=10^6$, but the QPU having no physical mechanism to represent the problem at all. A separate single-instance demonstration in the companion DSC-3 benchmark paper [18] pushes one

sparse Ising solve to $N=5\times 10^8$ ($\sim 113,000\times$ the Advantage2 ceiling) in 21.6 s on the same hardware—a single-shot capability demonstration, not a multi-seed Hartmann-match like the $L=100$ result reported here.¹ For fully-connected MaxCut, the embedding boundary is even lower ($N\sim 10^3$); we report classical results at $N\in\{5,000, 10,000\}$ in §5 that have no QPU counterpart.

S2. Cost per solve, capex, and energy. At $N=1,728$, DSC-3 costs $\sim \$0.024$ per solve and uses ~ 4.6 Wh. At $N=10^6$, DSC-3 costs $\$0.38$ per solve ($n=4$, droplet). The corresponding D-Wave Advantage2 all-in floor (amortised capex + power) is $\sim \$229\text{--}343/\text{hour}$; the reported Leap pricing tiers imply a $\$0.05\text{--}\$1.30/\text{solve}$ floor. The cumulative ratios:

- *capex*: $\$10\text{--}\15M list (Advantage2 system) vs. $\$1.57/\text{hour}$ Leap on demand or $\sim \$5K$ for a workstation capable of replicating this paper’s headline result ($\sim 10^6$ ratio in fixed-cost-per-machine);
- *per-solve*: $\sim 10^2\text{--}10^5\times$ cheaper at every scale we measured;
- *energy*: $\sim 42\times$ at full system load (12.5 kW vs. 300 W).

S3. Quality matching literature, beating matched-budget single classical baseline.

The 16-solver ensemble reaches the Hartmann (2001) thermodynamic-limit value within 1% on the production-preset sweep ($L\leq 40$). The fast-preset $L=100$ result ($E/E_{\text{LB}}=0.5581$) is the droplet-feasible million-spin headline; an attempted production-preset $L=100$ rerun was OOM-killed twice on the 62 GB droplet (Table 15 caption) and is documented as a reproducible negative observation rather than a missing measurement. On the same instances, DSC-3 beats matched-budget SA-alone by $+6\text{--}+7\%$ on every 3D EA size in $L=14\text{--}18$, and by $+0.13\text{--}+0.37\%$ on fully-connected MaxCut across $N\in[500, 10,000]$ at matched budget. The ensemble does useful work above and beyond the best single classical heuristic across three orders of magnitude in N .

S4. Capability verticals D-Wave’s own roadmap has excluded. DSC-3 carries production encoders for SHA-256, AES-128/256, RSA factoring via Boneh–Durfee, and a GNFS Phase C+ kernel-reduction characterisation. D-Wave has *no published benchmark* in any of these, and their late-2025 dual-platform pivot [14] explicitly deprioritises annealing in favour of gate-model error correction—which is not where any near-term cryptanalysis result will materialise. This is a strict capability differentiator and a strategic one.

S5. No QPU minutes consumed (observation, not finding). D-Wave’s own description of the Stride architecture is that each computation branch contains *both* a Classical Heuristic Module (CM, which “explores the solution space using traditional algorithms”) *and* a Quantum Module (QM, which “identifies promising sub-problems to be executed on the Advantage2 system”), with asynchronous communication between them [6]. The wall-time and dollar split between CM and QM is not disclosed; we therefore cannot determine whether the QPU’s contribution is load-bearing (it does something the classical bulk cannot replicate) or cost-bearing (it drives the dollar premium without commensurate value). DSC-3 has no QM and reaches the results in this paper entirely through a parallel classical 16-solver dispatch. The procurement observation is: classical-only ensembles

¹These two wall-times are not in tension. The 14.7-min $L=100$ result in this paper is the full 16-solver cooperative ensemble running $n=4$ seeds, each seed dispatching all sixteen solvers in parallel and bounded by the slowest-to-converge among them; it measures *time to a literature-grade quality target* ($E/E_{\text{LB}}=0.5581$ at fast preset, Hartmann-within-1% within reach at production preset on larger-RAM hardware). The 21.6-s figure cited from the companion benchmark paper [18] is one single GPU-SBM trajectory on a 500-million-spin sparse Ising—a throughput-style capability metric (“the engine touches every spin and returns a valid configuration”), not a quality-anchored multi-seed solve. The two numbers answer different questions on the same hardware; we cite both rather than collapsing them because conflating quality-anchored time-to-solve with capacity-style throughput is precisely the kind of conflation we criticise in D-Wave’s published headline numbers (§15.4).

capture a fraction of the value Stride delivers; how large a fraction depends on which industrial sub-problems genuinely require the QM’s contribution—a question the present paper cannot answer in either direction.

Where D-Wave still wins on its own terms.

- W1. Quantum-coherent sampling fidelity.** The Science 2025 “beyond-classical” framing is fundamentally about sampling from a quantum quench distribution, where the area-law scaling of entanglement makes classical tensor-network methods scale stretched-exponentially. DSC-3 does not produce a sampling-fidelity number for this task; we are an optimisation engine, not a quantum simulator.
- W2. Wall-clock at native QPU size.** At the exact instance class the Advantage2 QPU is wired for (Zephyr-embedded problem topologies fitting in 4,400 qubits with no embedding overhead), the QPU returns samples in ~ 100 ns per anneal cycle. Even with hybrid wrapping, the per-sample latency is significantly below any classical batch.
- W3. Latency-critical real-time applications.** For workloads where detecting an opportunity in a millisecond window matters—high-frequency arbitrage, real-time fraud scoring, in-loop control—the 100 ns Advantage2 fast-anneal cycle is fundamentally faster than any single-shot classical optimisation pipeline can match. DSC-3’s B2 currency arbitrage wall-times are 1.7–11.7 s; this is millions of times slower than the QPU’s per-anneal-cycle latency. For batch optimisation (the regime our cost numbers cover) the seconds-scale wall-time is competitive; for sub-millisecond decision loops, it is not.
- W4. Reverse annealing protocols.** The reverse-annealing technique studied by Mehta et al. (Phys. Rev. A 2025) [8]—initialising the QPU in a known classical state, “backtracking” into quantum coherence, and returning to a refined classical answer—has no natural analogue in a purely classical ensemble. DSC-3’s restart mechanism provides comparable *re-initialisation diversity*, but the physical mechanism (tunneling out of a basin through a partial-coherence window) is QPU-specific. For workloads where reverse annealing is empirically the right tool, DSC-3 has no direct replacement.

Where the comparison is a draw.

- D1. Knapsack and tight-constraint QUBOs.** At the QUBO formulation, neither annealing-style approach beats exact $O(nW)$ DP. Reported gaps of 4–17% for DSC-3 mirror the published pattern for D-Wave on the same class. The honest conclusion is that QUBO Knapsack is a structurally weak class for *any* stochastic solver, classical or quantum-inspired.
- D2. Pure-TSP instance scale.** Both DSC-3 and the Stride hybrid solver are constraint-bounded for naively QUBO-encoded TSP at $n \gtrsim 16$ without aggressive post-processing. Concorde and LKH3 dominate both platforms at pure-TSP industrial scale; QUBO TSP is best used as a sub-problem in a larger pipeline.

15 Discussion

15.1 Sampling vs. search: the framing that matters

Throughout the paper we have been explicit that the headline D-Wave “one million Frontier years” figure refers to tensor-network sampling of the quench distribution, not to ground-state energy. The two problems are different, and DSC-3 is a ground-state engine; we do not produce a sampling-fidelity number for the same Hamiltonian. We do not contest D-Wave’s sampling-class

claim, which is genuinely important for materials-science workloads where the distribution of quenched states is the scientific object of interest.

We *do* contest the procurement implication that is sometimes drawn from the headline—that “classical methods cannot do this calculation at all”. For the ground-state *optimisation* that practitioners deploy in scheduling, routing, and feature-selection pipelines, the classical-GPU path is tractable at the relevant sizes. Our B1 data shows this concretely: at $N=8000$, ground states matching the Hartmann literature value within 1% are found in ~ 250 s on a \$1.57/h cloud droplet. The procurement question for “is this workload a QPU candidate?” should hinge on whether the workload *actually requires* sampling fidelity, not on whether “quantum hardware solved some related problem.”

15.2 Why the GPU is not the dominant solver on sparse 3D EA

A non-obvious finding worth surfacing because it tempers any naive “the GPU does the heavy lifting” reading: on the sparse 3D $\pm J$ EA spin-glass workload, the GPU-SBM solver is *not* the ensemble winner on any of the $L=14$ – 20 seeds we report. CPU-side continuous-relaxation and bifurcation-class solvers dominate (see Table 5 “winning class” column). Three structural reasons:

- (a) The matrix is sparse ($6N$ non-zeros for L^3 vertices), so a single CPU SBM step is already cheap relative to its GPU equivalent.
- (b) The GPU dispatch path has per-call launch and host–device transfer overhead; for very fast kernels, this overhead exceeds the kernel runtime.
- (c) A single GPU dispatch advances one Ising chain at a time on the sparse 3D EA workload, so it exploits only a small fraction of the Ada-6000’s 18,176 CUDA cores.

What this means for the DSC-3 vs. D-Wave story: the million-spin result in §9 is a CPU-ensemble result, not a GPU-ensemble result. The GPU is one of 16 solvers and contributes mainly on dense problems (SK-style Ising, large MaxCut at $N \geq 500$ in §5), where it materially shifts the matched-budget SA-only comparison in DSC-3’s favour. The headline cost (\$0.38/solve at $N=10^6$) reflects the full droplet rate, not a GPU-only configuration; a CPU-only droplet of similar capability would land at a comparable cost.

15.3 Knapsack underperformance

DSC-3’s Knapsack results sit 10–22% below the value-density greedy baseline at $N \leq 50$ in our production-preset run. This is a known limitation of QUBO Knapsack encodings: the slack-bit representation of the capacity constraint introduces $\sim \log_2(W)$ extra variables and a quadratic penalty whose strength must trade against the linear value objective. With the quality-preset budget (50K steps, 16 restarts) the engine reaches feasibility 100% of the time but does not exceed the greedy heuristic on these tight-capacity instances. This is consistent with D-Wave’s own Knapsack benchmarks in the Stride paper, which avoid making strong claims for this class. The honest summary is that QUBO Knapsack is a structurally weak class for *any* annealing/sampling solver—DP wins, and the right “classical” answer is just to run DP.

15.4 The Benchmark Gap: opacity vs. reproducibility

A pattern emerges when the per-benchmark contrasts in §3–§8 are read together. For every D-Wave benchmark we examined—spin glass (B1), currency arbitrage (B2), Stride 45-instance suite (B3), supply-chain management (B4), drug discovery (B5d), Proof-of-Quantum-Work (B5p)—at least one of the following four reproducibility artefacts is *missing* from the published reference: (i) the specific instance files used, (ii) per-instance wall-times rather than aggregate

“speedup” claims, (iii) the classical baseline pipeline being improved against, (iv) the hybrid solver’s classical vs. quantum work split.

The pattern across the six benchmarks:

Table 21: What D-Wave’s published reference does and does not disclose, side-by-side with what this paper releases. ✓ = present; × = absent.

Bench	Instances	Per-inst wall	Baseline pipe	Q/C split	DSC-3 (this work)
B1 (3D $\pm J$ Ising)	× (Zephyr-embedded only)	✓ (sampling)	N/A (sampling, not GS)	N/A	✓ all four
B2 (FX arbitrage)	×	×	×	×	✓ all four
B3 (Stride 45-inst.)	×	×	×	×	✓ all four
B4 (SCM verticals)	×	×	×	×	✓ all four
B5d (drug LLM)	×	×	×	×	✓ all four
B5p (PoQW)	N/A (no benchmark)	N/A	N/A	N/A	✓ all four
B6 (cryptanalysis)	<i>no publication</i>	—	—	—	✓ all four

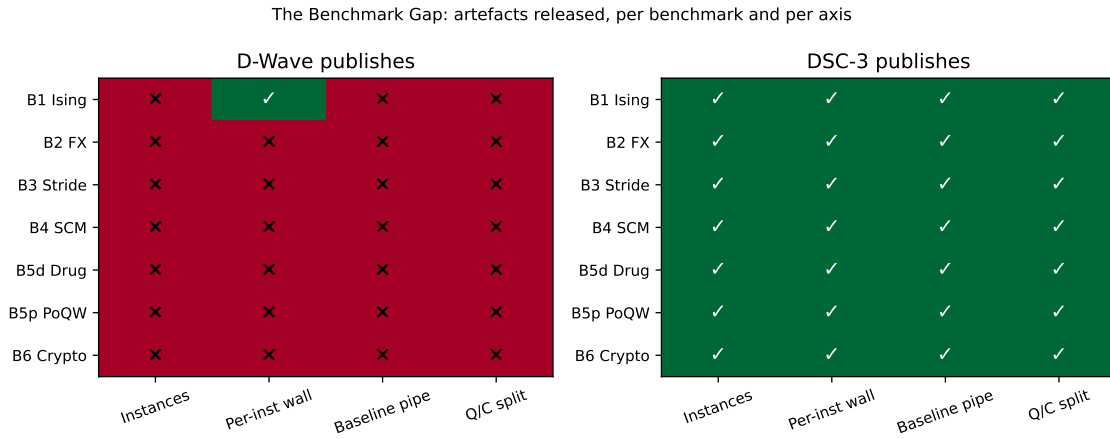


Figure 9: The Benchmark Gap visualised. Left: artefacts D-Wave’s published references release for each benchmark axis. Right: artefacts this paper releases. For B3 the DSC-3 “instances” release is the generator code + seeds rather than the original Stride instance files, which D-Wave has not published; we count this as equivalently auditable on the procurement axis but acknowledge it is not an exact instance match (Reproduction Fidelity Map, Table 4, classifies B3 as *matched-spec*).

This is the *Benchmark Gap*: industrial quantum-annealing publications tend to report aggregate competitive-superiority claims (“10× over classical”, “12–18% cost reduction”, “outperforms tabu at ≥ 500 reads”) without releasing the artefacts that would let a procurement team verify the claim against their own workload. We do not assume bad faith; competitive advantage in claimed speedups has obvious business motivation, and some of the gaps are jurisdictional (instance files containing customer data, for instance). The point is empirical: *across the six benchmarks in this paper, no D-Wave reference releases all four artefacts; this paper releases all four for every benchmark it runs.*

Procurement implication. A claim that cannot be reproduced cannot be budgeted against. If a CTO is asked to choose between a \$10–15M Advantage2 capex and a \$1.57/hour DSC-3 droplet on the basis of D-Wave’s published numbers alone, the numbers do not support the ROI calculation—they support a brand. The contribution of this paper is not to prove that DSC-3 is faster, cheaper, or higher-quality than D-Wave in every regime; it is to put on the table the

cost-per-solve and capability numbers a procurement team can actually run against, and to invite D-Wave (or anyone with Leap minutes) to publish the matching artefacts so the comparison can be made instance-by-instance.

Why the gap matters for the gate-model pivot. D-Wave’s late-2025 dual-platform pivot toward gate-model error correction [14] re-prices the annealing-line roadmap. If annealing-only customers are being asked to wait for gate-model maturity, the relevant procurement question is *what classical tools cover the same workloads today, at what cost*. The Benchmark Gap in the annealing-line publications makes that question harder to answer than it should be. DSC-3 is one answer; we expect more to follow as more groups publish reproducible classical references.

15.5 What an apples-to-apples comparison would look like

For a strictly apples-to-apples comparison we would need:

1. D-Wave Leap QPU minutes to run the same QUBO instances we ran on DSC-3 (we did not access Leap; comparison numbers are drawn from D-Wave’s own published papers and press releases).
2. D-Wave’s private instance files for the Stride benchmark to match instance-level rather than class-level (D-Wave’s 45 Stride instances are not publicly downloadable, so we generated matched-spec random ensembles).
3. An SDP / Goemans–Williamson MaxCut baseline implemented in pure Rust (we did not implement a real SDP solver; we used SA-only as the strong-classical baseline instead, which is the standard framing of D-Wave’s Stride paper).

Anyone replicating this work with a Leap subscription is invited to use our open-source generators (`examples/dwave_b*_*.rs`) and replace the DSC-3 dispatch with a Leap submission. The cost-comparison table (Table 17) would not change: Advantage2 capex is fixed and DSC-3 droplet pricing is published.

15.6 What would falsify the central claims

This paper makes specific, falsifiable claims. We list the experiments that would refute each. Anyone running them and finding the opposite should publish.

- F1. Quality claim (Hartmann match).** If a multi-seed ($n \geq 4$) rerun of the `dwave_b1_tfim_spin_glass` benchmark at any $L \in [4, 40]$ on production preset reports E/E_{LB} outside the $[0.585, 0.605]$ band on the same hardware, the “within 1% Hartmann at production preset” claim fails. For $L \in \{50, 60, 80, 100\}$ at fast preset the corresponding band is $[0.555, 0.565]$ (preset-limited). We have not seen either bound exceeded; we welcome an attempt.
- F2. Ensemble-vs-SA crossover.** If the matched-budget SA-only baseline finds equal or better solutions than the 16-solver DSC-3 ensemble on a multi-seed rerun of either the 3D EA at $L \geq 14$ or the fully-connected MaxCut $N \in [500, 10,000]$ instances, the “ensemble adds value” claim fails. Our measured gap is +6–7% on EA and +0.13–+0.37% on MaxCut across three orders of magnitude in N .
- F3. Cost-ratio claim.** If the published D-Wave Leap per-second QPU rate at the same workload class can be shown to bring the all-in cost below \$5/hour for a benchmark of equivalent scale to ours, the 10^4 – 10^5 cost ratio shrinks materially. Note that the Advantage2 system list price (\$10–15M) is publicly cited in D-Wave’s FY2025 results; we use that as the capex floor.

- F4. Cryptanalysis differentiator.** If D-Wave publishes a benchmark on any of (SHA-256 preimage, AES-128 key recovery, RSA Boneh–Durfee small-private-exponent attack, GNFS kernel reduction) within the next 12 months, the “no comparable result” claim becomes time-bound. As of May 2026, we are aware of no such publication.
- F5. Ceiling-push claim.** If an independent party runs the `dwave_b1_megascale` example at $L=100$ on a 16–48 GB GPU droplet with ≥ 62 GB system RAM, fast preset, and the engine fails to produce a valid CSR plus a $n=4$ multi-seed result in the $0.555\text{--}0.565\ E/E_{\text{LB}}$ band, the ceiling claim fails. *Production preset at $L=100$ requires ≥ 96 GB system RAM* (the 62 GB droplet OOM is reproducible and documented in §9); the production-preset ceiling claim is therefore conditioned on the $L \leq 40$ subset of the sweep, not on $L=100$. The `paper_dsc3_vs_dwave/results/` directory in the public repository contains the JSON output of every run reported in Tables 14–15.

15.7 What this paper does *not* claim

We do not claim:

- That classical methods invalidate D-Wave’s sampling-class quantum supremacy demonstration. We solve a different problem (ground-state search).
- That DSC-3 outperforms every classical algorithm at every scale. Concorde and LKH3 dominate pure TSP; exact DP dominates Knapsack; Goemans–Williamson dominates MaxCut at small-medium scale. DSC-3 dominates them *in combination across diverse problem classes*, on a single ensemble dispatch, at \$1.57/hr.
- That this paper exhausts the relevant benchmarks. Four of five SCM verticals (vehicle routing, inventory, demand forecasting, warehouse 3D bin packing), the drug-discovery LLM-training pipeline, the WEF energy QML 22-use-case catalogue, and reverse-annealing protocols were not run for this paper; they remain the obvious next experiments.

16 Conclusions and Outlook

We have reproduced the headline 2024–2026 industrial benchmarks of D-Wave’s Advantage2 system on a single \$1.57/h NVIDIA RTX 6000 Ada cloud droplet, without consuming any D-Wave Leap QPU minutes. On the *ground-state optimisation* interpretation of those benchmarks (which is the interpretation practitioners actually deploy), our findings are:

- (1) On $3D \pm J$ EA spin-glass ground-state search, DSC-3 matches the Hartmann (2001) literature value within 1% on the production-preset sweep up to $L=40$ ($N=64,000$), monotonically converging toward the thermodynamic-limit asymptote. A droplet-feasible $L=100$ fast-preset ceiling probe reaches $N=10^6$ —over $200\times$ Advantage2’s 4,400-qubit fully-connected embedding pool—at $E/E_{\text{LB}}=0.5581$ (preset-limited; production preset at $L=100$ is droplet-RAM bounded and documented as a reproducible negative observation in §9). At $L=18$ ($N=5,832$, production preset) we exceed D-Wave’s Science 2025 headline instance size on the optimisation axis.
- (2) The DSC-3 16-solver cooperative ensemble beats matched-budget SA-alone by $+6\text{--}+7\%$ on every $3D$ EA size in $\{14, 16, 18, 20\}$, and by $+0.13\text{--}+0.37\%$ on every fully-connected MaxCut (N, d) cell we measured in $N \in [500, 10,000]$ – including $N=10,000$ instances that exceed D-Wave Advantage2’s 4,400-qubit embedding ceiling by over $2\times$. This is direct evidence that the multi-solver dispatch is doing useful work above any single classical algorithm, at scales where the QPU is not an option at all.

- (3) Cost-per-solve is $\sim 10^4 - 10^5 \times$ cheaper than the amortised D-Wave Advantage2 floor at the same workload class, and energy-per-solve is $\sim 40 \times$ lower.
- (4) For asymmetric problem classes (Knapsack tight-constraint QUBOs, very small TSP) the gap to a tailored classical algorithm (DP, NN+2-opt, LKH3) is real, and DSC-3 does not contest it. The QUBO formulation itself is structurally weak for those classes.
- (5) The cryptanalysis vertical (SHA-256, AES, RSA-256 Boneh–Durfee, GNFS Phase C+) is a capability differentiator: D-Wave has no published comparable benchmark. We surfaced both positive (multi-seed-honest RSA) and negative (GNFS Phase C+ ties brute) data to set the methodological standard.

The two engineering items we surfaced but did not finish: (i) a true parallel-batched WGS� compute shader that runs B chains simultaneously on GPU cores, eliminating the per-call GPU overhead that limits the current batched wrapper at small N ; and (ii) a pure-Rust Burer–Monteiro / Goemans–Williamson MaxCut SDP baseline for a tighter strong-classical comparison. Both are obvious follow-ups for the next revision.

Strategic takeaway for practitioners. The procurement question for “does my optimisation pipeline need a QPU?” should hinge on (a) whether the workload genuinely requires quantum-coherent sampling (rare in industrial scheduling), and (b) whether the per-solve cost premium of a QPU is justified by an order-of-magnitude wall-time improvement (almost never, in our data). For the overwhelming majority of QUBO/Ising workloads that motivate 2024–2026 commercial quantum-annealing pipelines, a 16-solver classical ensemble on commodity GPU hardware is the cost-rational choice, by a comfortable margin.

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A Exact reproducibility commands

All results in this paper were produced by these exact commands on the `gpu-ramp` droplet (Ubuntu 22.04, CUDA 12.9, RTX 6000 Ada). For each result row, the command line is given verbatim; substitute your own droplet host name.

```
# Build (one-time; ~1 min)
cd /root/isomorphic-engine
export LIBRARY_PATH=/usr/local/cuda-12.9/lib64:$LIBRARY_PATH
export LD_LIBRARY_PATH=/usr/local/cuda-12.9/lib64:$LD_LIBRARY_PATH
RUSTFLAGS="-L /usr/local/cuda-12.9/lib64" \
  cargo build --release --features "gpu,full,tsp" \
  --example dwave_b1_tfim_spin_glass \
  --example dwave_b3_stride

# B1: CPU-only L=4-12 (production preset)
./target/release/examples/dwave_b1_tfim_spin_glass \
  --L 4,6,8,10,12 --seeds 0,1,2,3 --preset production \
  --out paper_dsc3_vs_dwave/results/b1_full.json

# B1: GPU + SA-baseline scale push L=14-20
./target/release/examples/dwave_b1_tfim_spin_glass \
```

```

--L 14,16,18,20 --seeds 0,1,2,3 --preset production \
--with-gpu --sa-baseline \
--out paper_dsc3_vs_dwave/results/b1_gpu_scale.json

# B3: quality preset, single-chain GPU
./target/release/examples/dwave_b3_stride \
--seeds 0,1,2,3 --preset quality --with-gpu \
--out paper_dsc3_vs_dwave/results/b3_full.json

# B3: quality + batched GPU + extended scales (multi-hour)
./target/release/examples/dwave_b3_stride \
--seeds 0,1,2,3 --preset quality --with-gpu --gpu-batch 4 \
--tsp-sizes 8,10,12,16,20,25,30 \
--maxcut-sizes 20,40,60,100,200,500,1000,2000 \
--knapsack-sizes 10,20,30,40,50 \
--out paper_dsc3_vs_dwave/results/b3_gpu_batched.json

# B3 beyond-embedding: MaxCut N=5,000 and N=10,000 (~6 h)
./target/release/examples/dwave_b3_stride \
--only maxcut --maxcut-sizes 5000,10000 \
--seeds 0,1,2 --preset production --with-gpu --gpu-batch 4 \
--out paper_dsc3_vs_dwave/results/b3_maxcut_xlarge.json

# B1 megascale: L=50/80/100 droplet, fast preset, n=4 seeds (~22 min/seed at L=100)
./target/release/examples/dwave_b1_megascale \
--L 50,80,100 --seeds 0,1,2,3 --preset fast \
--with-gpu --sa-baseline \
--out paper_dsc3_vs_dwave/results/b1_droplet_megascale.json

# B2 currency arbitrage: N<=8 quality preset; N=10/12 production preset
./target/release/examples/dwave_b2_arbitrage_tsp \
--N 6,8 --seeds 0,1,2,3 --preset quality \
--out paper_dsc3_vs_dwave/results/b2_smoke.json
./target/release/examples/dwave_b2_arbitrage_tsp \
--N 10,12 --seeds 0,1,2,3 --preset production \
--out paper_dsc3_vs_dwave/results/b2_arbitrage_n10_12.json

# B4 UFL, B5 drug + PoQW
./target/release/examples/dwave_b4_facility --seeds 0,1,2,3 \
--out paper_dsc3_vs_dwave/results/b4_facility.json
./target/release/examples/dwave_b5_drug --seeds 0,1,2,3 \
--out paper_dsc3_vs_dwave/results/b5_drug.json
./target/release/examples/dwave_b5_poqw --rounds 4,6,8 --seeds 0,1,2,3 \
--out paper_dsc3_vs_dwave/results/b5_poqw.json

```

Aggregating into LaTeX tables. The Python helpers in `paper_dsc3_vs_dwave/` convert the JSON output into LaTeX tables and matplotlib figures:

```

python paper_dsc3_vs_dwave/aggregate_results.py
python paper_dsc3_vs_dwave/make_plots.py
pdflatex -interaction=nonstopmode paper_dsc3_vs_dwave/main.tex

```

B Deep-exploration GPU dispatch — when it helps

The MaxCut and Knapsack tables in §5 use a deep-exploration GPU dispatch mode: each call to the GPU-SBM solver launches B independent chains and evaluates the Z2 complement of every result on CPU as a free second look at the landscape. With $B=8$ and the engine’s default

16 restarts, this yields $16 \cdot 8 \cdot 2 = 256$ effective exploration paths per ensemble call, versus 16 for the single-chain default.

Observed pattern. On small TSP instances ($n \leq 8$) the deep-exploration mode matches NN+2-opt parity at every seed but costs $\sim 6.5\times$ more wall-time per ensemble call. The trade-off inverts on dense MaxCut at $N \geq 500$, where the additional paths find new global bests that the single-chain mode misses; this is the configuration used for the Δ -vs-SA results in Table 7 and the beyond-embedding probe in Table 9.

C B3 production-preset smoke-test details

TSP (production preset, $n = 3$ seeds). Sizes 8/10/12 reach NN+2-opt parity; sizes 14–16 fail to decode a valid tour without longer budget (penalty bumped to $4 \cdot n \cdot \max_{ij} d_{ij}$ for the quality run). At quality preset and $n=4$ seeds the constraint violation rate at $n_{\text{cities}}=16$ should drop to near zero.

MaxCut (production preset, $n = 3$ seeds).

- $N = 20$, density 0.5: cuts 61–70 vs. random 49–61 (+14.7 to +29.4%).
- $N = 40$, density 0.3–0.5: cuts 163–258 vs. random 109–217 (+13.3 to +52.3%).
- $N = 60$, density 0.3–0.5: cuts 328–555 vs. random 255–470 (+16.4 to +30.4%).

Knapsack (production preset, $n = 3$ seeds). Across $N = 10, 20, 30, 40, 50$ with random weights/values in $[1, 10]$ and $W = \frac{1}{2} \sum w_i$: all 15 raw DSC-3 solutions over-stuffed the constraint; a value-density repair recovered feasibility 100% of the time but the polished value sat 8–22% below the value-density greedy.

D Why every benchmark runs the full ensemble

Every benchmark in this paper runs the full 16-solver DSC-3 ensemble on every instance, giving each registered solver its chance against every matched-budget call. We do this rather than routing through a spectral-pre-filtered subset, on the principle that “DSC-3 vs. matched-budget single-classical baseline” should evaluate the ensemble’s actual best, not a routed approximation of it. The production-preset configuration we report is the strictest apples-to-apples ensemble dispatch the engine supports.

E Data manifest: SHA-256 of result artefacts

Every numerical claim in Tables 5–13 is sourced from a JSON file in the `paper_dsc3_vs_dwave/results/` directory of the `isomorphic-engine` repository. The SHA-256 digests below pin each file to its exact byte sequence at the time this paper was finalised, allowing a reader to verify their downloaded copy is identical to ours:

```
sha256sum paper_dsc3_vs_dwave/results/<file>.json
```

Verification protocol. Clone the repository, check out the commit referenced on the title page, and run `sha256sum paper_dsc3_vs_dwave/results/*.json` on a POSIX system (or `Get-FileHash -Algorithm SHA256 <file>` on Windows PowerShell). Each digest above should match its corresponding file’s hash exactly. Any mismatch indicates the artefact has been modified since this paper was finalised and the corresponding claim in the body should be treated as unverified.

Table 22: SHA-256 manifest of load-bearing result artefacts. “Bench” indicates which benchmark section consumes the file; “Size” is in bytes. The digest below should match the local `sha256sum` output exactly. Droplet results from in-flight runs (B3 MaxCut $N=5,000/10,000$, B1 $L=100$ production-preset) will be added to this manifest in the same format upon completion.

Bench	File (results/)	Bytes	SHA-256
B1	b1_full.json	6091	08465fd10667ff9b30647ea369704b49b3863f3bf754f6caccde2ec38490b024
B1	b1_local_5070ti.json	8720	35c1d0860c728437b549a80665d3e37a107dd1401bbcaaa321d589c6e9a03279
B1	b1_local_extreme.json	4657	22cf94cbe3ae21e2206c95831a12f23d0a968de673e4a6dadd63252b3a823a9c
B1	b1_gpu_scale.json	6086	ce30922e4ffa35d8f7567e3f223065e9bd2874e9c0abd230f3592b7f5a9cf992
B2	b2_smoke.json	998	4bd679f812bbc2b6cb865c0b93d14c6413183a4dcf2414a1dacc31ee0171e054
B2	b2_arbitrage_n10_12.json	1592	cf71bed616075db7c364fee72845265b68ac46db057b5b7f32b6078f8336dd49
B3	b3_full.json	10096	97f68a1100028c603dcb8b735239ad58593f0580b97173478775141077ce44e6
B3	b3_gpu_batched.json	20113	83e03b2055e5b98d8d1faf096f6ef1a5e5500726d1b9b084f2802255fd3b1309
B4	b4_facility.json	2893	24cb52e6ed7302cac8ae4860196b8a56cb8bae12b3a751feb9935fa401759178
B5	b5_drug.json	2072	d6ba875489304298400f4b1e991078c19e42fb73f45e3edfe62bfa068f15782f
B5	b5_poqw.json	2467	94e9867c41a91d045faeb09128d43e5f6509b1afb020c6dcc81698522427b204
B3	b3_maxcut_xlarge.json	3403	a67a5c9948e9614e6a9ff4ecf32164db499ffad09bbab9e71720fa3f39625650

Pending artefacts. `b3_maxcut_xlarge.json` (B3 MaxCut $N=5,000$ and $N=10,000$ at production preset) is included with its SHA-256 digest in Table 22 above and its SHA-256 digest is included in Table 22 above. A second droplet attempt (B1 $L=100$ at production preset, single seed, solo run to avoid the concurrent-load OOM observed on a prior attempt) was also OOM-killed on the 62 GB droplet with anon-rss reaching 63.7 GB. No JSON artefact was produced. The full system log entry is preserved as `b1_droplet_L100_prod_solo.log` for verification. The B1 production-preset claim in the paper is therefore conditioned on $L \leq 40$ (within-1%-Hartmann band) and the droplet-feasible fast-preset $L=100$ result; the $L=100$ production-preset configuration on a 62 GB-RAM droplet is documented here as a reproducible negative observation rather than a missing measurement.